Certifying neural eigenpairs and degenerate eigenspaces of the Schrödinger equation: a self-verifying *a posteriori* protocol and its implementation

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Abstract

Neural solvers of the stationary Schrödinger equation are evaluated against a reference solution that new problems lack. We present a computable *a posteriori* certificate for neural eigenpairs of the discretized operator: from an approximate eigenfunction it returns a two-sided eigenvalue enclosure, a bound on the angle to the true eigenfunction, and a verified state index, using only the approximation itself. The hypotheses of the Temple and Davis–Kahan bounds are *verified* rather than assumed, by counting the spectrum below a shift with Sturm sequences in one dimension and with the Sylvester inertia of a sparse LU factorization in two; the one-dimensional count, and the two-dimensional one for separable potentials, are re-derived in directed rounding, hence theorems about the stored matrix. A degenerate level has no single eigenfunction to certify, so we extend the construction to invariant subspaces: the multiplicity is *measured*, cluster eigenvalues are bracketed, and a principal-angle bound certifies the subspace spanned by several network outputs. On six potentials, all 39 enclosures contain an independently computed discrete eigenvalue; a degenerate state with residual 4×10^{-12} is refused individually and certified as a two-dimensional subspace; independently trained pairs that span it are separated from collapsed ones. A variational bound on the interpolant carries one side of the guarantee to the continuous operator.

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Keywords: Schrödinger equation, neural network eigen solver, a posteriori error bound, Temple bound, Sylvester inertia, invariant subspaces, verification

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Program summary

Program title: `nn-schrodinger-certify`
CPC Library link to program files: (to be added by Technical Editor)
Developer's repository link:
`https://github.com/YuhaoPeng1103/PINN_equation`
Licensing provisions: MIT license (MIT)
Programming language: Python 3.11 (NumPy, SciPy; PyTorch for training)
Supplementary material: none
Nature of problem (approx. 50–250 words): Neural solvers of the stationary Schrödinger equation are evaluated against a reference solution that a new problem does not have. A small residual does not bound the eigenfunction error, and at a degenerate level no single eigenfunction is defined, so the usual comparison is ill-posed as well as unavailable: what is wanted is a statement computable from the approximation alone.
Solution method (approx. 50–250 words): Given an approximate eigenfunction, or an m-dimensional trial subspace, on a finite-difference grid, the code returns a two-sided enclosure of the discrete eigenvalue, a bound on the angle to the exact eigenfunction or invariant subspace, and a verified state index. The hypotheses of the Kato–Temple and Davis–Kahan bounds are decided by counting the spectrum with Sturm sequences in one dimension and with the Sylvester inertia of a no-pivot sparse LU in two, guarded against silent pivoting and against a resolution floor; the one-dimensional count, and the two-dimensional one for separable potentials, are re-derived in directed-rounding interval arithmetic and refuse when a pivot or a summed pair is unresolved. A variational bound evaluated on the P1/Q1 interpolant of a sampled eigenfunction bounds the continuous eigenvalue from above without a reference solution, and a certificate-selected Dirichlet sine filter and a transport inequality extend the bound to the raw network output.
Additional comments including restrictions and unusual features (approx. 50–250 words): Includes the drivers that regenerate every table and figure of the article and a 143-test suite. A count costs 0.03 s at 81×81 and 0.4 ms at $N = 2001$, or 5 ms in one dimension with directed rounding; a full certificate costs 0.07 s at $N = 2001$ and 0.2 s at 81×81. Restriction: the operator must be assemblable, so the high-dimensional regime is out of reach; the anharmonic potential is counted in guarded floating point because the general two-dimensional interval count is measured but not used; the resolution floor is a declared margin, not a theorem.
References: none

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1. Introduction

Neural representations of quantum wavefunctions are now used both as variational ansätze for many-body problems [1, 2, 3] and as mesh-free solvers for the stationary Schrödinger equation [4, 5]. In the second setting a network ψ_θ is trained so that $H_h\psi_\theta \approx E\psi_\theta$ in a discrete L^2 sense, and the paper reports an energy error and a wavefunction overlap against a reference solution.

This evaluation scheme has a structural weakness. The reference is either an analytic solution—available only for textbook potentials—or the output of a classical solver on the same grid. In the second case the comparison measures agreement between two approximations, not accuracy, and it silently absorbs the discretization error of the grid into the "method error". More importantly, in a genuinely new problem there is no reference at all, and the only quantities available are those the approximation itself provides. A residual is such a quantity, but making it small is itself delicate—the gradient pathologies of residual training are well documented [6]—and even a small residual is not an error: a residual of 10^{-2} may correspond to an energy error of 10^{-6} or to a wavefunction that is entirely the wrong eigenstate. Both directions occur in the runs reported below. A network whose energy was accurate to seven digits ($|\eta - \lambda_h| = 2.6 \times 10^{-7}$ on the infinite-well grid) carried a residual of 0.53, so that the certificate could prove only an angle bound of 0.56 rad and an enclosure six orders of magnitude wider than the true error (Section 7.4). Conversely, a network trained with a scale-invariant loss returned a field whose residual norm was 2.9×10^{-4} while its L^2 distance from the eigenfunction of unit norm was 0.99: the residual had been made small by shrinking the field (Appendix A.1).

The numerical analysis literature solved the corresponding problem for finite element and spectral methods long ago: eigenvalue enclosures follow from variational bounds (Weinstein, Kato–Temple, Lehmann–Goerisch) together with a verified count of the spectrum in an interval [7, 8], and eigenvector bounds follow from the $\sin\theta$ theorems of Davis and Kahan [9]. Verified-computing packages routinely return such enclosures with directed rounding. What has not been done, to our knowledge, is to connect this machinery to neural eigenpairs, where the "approximation" is a mesh-free function that must first be sampled on a grid, where the hypothesis of the classical theorems (an interval containing exactly one eigenvalue) is not available a priori, and where the training procedure itself can leak the reference solution into the object being evaluated.

This paper closes that gap with a protocol and a reference implementation. The ingredients are classical; the contribution is that they are assembled into something that can be run on a neural eigenpair and that verifies its own hypotheses. Specifically:

1. **A computable certificate** (Section 2) that takes an approximate eigenfunction ψ on a grid and returns (i) an interval $[a, b]$ that is *verified* to contain exactly one eigenvalue of the discrete Hamiltonian H_h, (ii) a two-sided enclosure of that eigenvalue from the Kato–Temple bound, and (iii) an upper bound on the angle between ψ and the exact discrete eigenfunction from the Davis–Kahan theorem. The verification of (i) uses Sturm sequences in one dimension and the Sylvester inertia of a no-pivot sparse LU factorization in two; it never uses a reference eigenvalue.
2. **Two numerical safeguards** that the verification needs in floating point: an explicit check that the factorization performed no pivoting, and a *resolution floor* below which an interval is declared unverifiable. The second is what makes the single-state protocol refuse a degenerate state instead of certifying it with a spuriously narrow interval (Section 2.5).
3. **A cluster certificate** (Section 4) for the case the single-state certificate must refuse. At a degenerate level no eigenfunction is defined by the problem, so we certify the invariant subspace instead: the multiplicity is measured by the same verified count, each cluster eigenvalue is bracketed to within the residual of its Ritz pair, and a principal-angle bound (Theorem 3) certifies the subspace spanned by a set of network outputs without separating their Ritz values. The construction also yields the failure modes that matter in practice—a subspace that missed a state is refused because its dimension does not match the count, and a set of independent training runs that collapsed onto the same representative is either refused (when the only second direction they contain is the difference of two nearly identical outputs) or accepted with a bound orders of magnitude weaker than a genuine pair's.
4. **A protocol** (Section 3) that uses the certificate in four ways: to remove leakage from the training residual by evaluating the energy at the Rayleigh quotient of the current iterate, to select the truncation order of a Dirichlet spectral filter a posteriori without any reference, and to carry the certified angle bound back to the unfiltered network output through an elementary transport inequality, and to decide which sets of

independent outputs actually span a degenerate level (Section 7.7).

5. **A directed-rounding count** (Section 6): the one-dimensional Sturm count, and the two-dimensional one for separable potentials, are recomputed in interval arithmetic with outward rounding, so that a verified count is a theorem about the stored matrix rather than a guarded floating-point estimate. An unresolved pivot or an unresolved sum is *refused*, with its position, never rounded to an integer; bisection on the certified count returns certified brackets for discrete eigenvalues in one and two dimensions.
6. **A variational bound on the continuous operator** (Section 6.4): any conforming trial function is a witness that $\lambda_1(\mathcal{H})$ does not exceed its Rayleigh quotient, and the P1/Q1 interpolant of a sampled eigenfunction—including a network's—is conforming, so the bound on $\mathcal{H}$ costs one quadrature and no reference solution, no grid operator and no interpolation constant. The matching lower bound needs a comparison principle; we obtain it for the infinite well and the box and show it to fail for the finite well, whose certified discrete energy can lie above the continuous one.
7. **An empirical audit** (Section 7) on six potentials in one and two dimensions, three seeds each, including a degeneracy experiment, a controlled splitting that locates the boundary between state and subspace certification, a leakage comparison, and a separation of network error from discretization error.

We state clearly what is *not* claimed. We do not propose new spectral inequalities: Weinstein's, Temple's and Davis–Kahan's bounds are classical, and the cluster statements of Section 4 are rearrangements of the same two arguments applied to an invariant subspace rather than to a vector; what is new is the construction that makes them *decidable* for a degenerate level, where the single-state hypothesis cannot hold. The two-sided enclosures of Sections 2–4 refer to the finite-difference operator H_h, and the bridge to the continuum in Section 6 is one-sided: an upper bound for every potential, a lower bound only where a comparison principle exists. Interval arithmetic does not reach the general two-dimensional count either—the separable case is exact, while the interval elimination refuses on most shifts, and we report that measurement rather than route around it. Finally, we do not claim that the certificate is tight. It is not, for non-smooth potentials; we report the ratio between certified width and true error rather than hiding it.

The intended use of the protocol is therefore not to replace an eigensolver. On every problem in this paper a classical solver produces the eigenpair in a fraction of the time, and the certificate applied to that eigenpair is 10^7–10^{10} times narrower than the one applied to a trained network, except on the two infinite wells, where the filtered network is itself the exact sine mode and the two certificates coincide at the round-off floor. The intended use is to answer, for a network that has already been trained, the question "how wrong can this be?", and to make that answer an auditable number in a table.

2. The certificate

2.1. Setting and notation

Let $\Omega = (0, L)^d \subset \mathbb{R}^d$, $d \in \{1, 2\}$, and let $x_i = a + ih$, $i = 0, \dots, n-1$, $h = L/(n-1)$, be a uniform grid with n points per dimension. We denote by $I = \{1, \dots, n-2\}^d$ the interior nodes and by $H_h : \mathbb{R}^{|I|} \to \mathbb{R}^{|I|}$ the five-point (three-point in 1D) Dirichlet discretization

$$H_h\psi = -\tfrac{1}{2}\Delta_h\psi + V\psi, \qquad \Delta_h\psi = \sum_{k=1}^{d} \frac{\psi_{i-e_k} - 2\psi_i + \psi_{i+e_k}}{h^2}, \tag{1}$$

with ψ extended by zero on ∂I (hard Dirichlet data). The discrete inner product and norm are

$$\langle u, v\rangle_h = h^d \sum_{i \in I} u_i v_i, \qquad \|u\|_h = \langle u, u\rangle_h^{1/2}. \tag{2}$$

H_h is symmetric, so its spectrum $\sigma(H_h) = \{\lambda_0 \le \lambda_1 \le \cdots\}$ is real and may be ordered. It is *not* positive definite in general—the finite well used below has $V_0 = 50$ and a ground energy of -48.97—and nothing that follows uses positivity: the count, the Temple bound and the angle bound need only symmetry and an isolating interval. All statements in this section refer to H_h, never to the continuous operator; Section 7.8 quantifies the difference.

Definition 1 (Approximate eigenpair)**.** *For a nonzero $\psi \in \mathbb{R}^{|I|}$ (extended by zero), define the discrete Rayleigh quotient and the residual*

$$\eta(\psi) = \frac{\langle \psi, H_h\psi\rangle_h}{\langle \psi, \psi\rangle_h}, \qquad r(\psi) = H_h\psi - \eta(\psi)\psi, \qquad \rho = \|r\|_h. \tag{3}$$

We call $(\psi, \eta(\psi))$ an approximate eigenpair with residual norm ρ.

The choice $\eta = \eta(\psi)$ is not cosmetic: it is the value of E that minimizes $\|H_h\psi - E\psi\|_h$, and it is the same object that enters the bounds below. Using a different energy—for instance a learned scalar or a reference value—in the residual produces a quantity that is not the certificate's residual, a point we return to in Section 3.1.

Lemma 1 (Weinstein [10])**.** $\operatorname{dist}\big(\eta, \sigma(H_h)\big) \le \rho$.

Proof. If $\eta \notin \sigma(H_h)$ then $(H_h - \eta)^{-1}$ exists with norm $1/\operatorname{dist}(\eta, \sigma(H_h))$, and $\psi = (H_h - \eta)^{-1} r$ gives $1 = \|\psi\|_h \le \rho/\operatorname{dist}(\eta, \sigma(H_h))$. □

Lemma 1 needs no hypothesis and no reference solution: it says only that *some* eigenvalue lies within ρ of η. It does not say which one, and it does not bound the eigenfunction. To obtain a two-sided enclosure of a *specific* eigenvalue and an angle bound for its eigenfunction, one needs an interval known to contain exactly one eigenvalue. The rest of this section is about obtaining that knowledge.

2.2. Verified counting of the spectrum

For $\sigma \in \mathbb{R}$ define

$$N(\sigma) = \#\{\lambda \in \sigma(H_h) : \lambda < \sigma\}. \tag{4}$$

If $a < \eta < b$ and $N(b) - N(a) = 1$, then (a, b) contains exactly one eigenvalue, and that eigenvalue is the k-th one with $k = N(a)$.

One dimension. $H_h - \sigma I$ is symmetric tridiagonal with diagonal $d_i = V(x_i) - \sigma + 1/h^2$ and constant off-diagonal $-1/(2h^2)$. The Sturm sequence

$$p_1 = d_1, \qquad p_i = d_i - \left(\frac{1}{2h^2}\right)^2 \frac{1}{p_{i-1}}, \qquad i = 2, \dots, |I|, \tag{5}$$

is the pivot sequence of the $\mathrm{LDL}^{\mathsf{T}}$ factorization, and by Sylvester's law of inertia $N(\sigma)$ equals the number of negative p_i. The matrix $H_h - \sigma I$ is singular if and only if some $p_i = 0$, a case the retry logic described below excludes. The recurrence is $O(|I|)$; a count is cheaper than a single back-substitution.

Two dimensions. The Sylvester law of inertia states that for a symmetric matrix A the triple (n_+, n_-, n_0) of positive, negative and zero eigenvalues is invariant under congruence [11]. If $A = LU$ with L unit lower triangular and no row or column exchanges, then $U = DL^\top$ with $D = \mathrm{diag}(U_{ii})$ and $A = LDL^\top$, hence

$$N(\sigma) = \#\{i : U_{ii} < 0\} \qquad \text{for } A = H_h - \sigma I. \tag{6}$$

We compute the factorization with SciPy's `splu`, called as

```
splu(A, permc_spec="NATURAL", diag_pivot_thresh=0.0,
                              SymmetricMode),
```

which suppresses both column permutations and threshold pivoting. Because a *silent* pivot in a library routine would invalidate (6), the implementation checks three conditions before using the count:

1. the returned row and column permutations are both the identity;
2. $\min_i |U_{ii}| > 10^{-13}\, s$, where $s = \max\{1,\ \tfrac{1}{2}(2/h_x^2 + 2/h_y^2) + \max|V|\}$ is the natural scale of H_h (a degenerate pivot makes the sign of U_{ii} meaningless);
3. the shift is not within the resolution floor of a spectral gap (Section 2.5).

If (i) or (ii) fails, the shift is perturbed by $10^{-11} s \cdot 10^k$ and the factorization repeated, up to six times; if all attempts fail the count is reported as unavailable rather than guessed. For the 81×81 grid used below ($|I| = 6241$, bandwidth 79) one factorization takes about 0.03 s on a single CPU core at $N = 81$ (and 0.13 s at $N = 121$), which makes the four to six counts per certificate negligible next to training.

2.3. Constructing an isolating interval

The approximate spectrum produced by a shift-invert Lanczos iteration, computed with `ARPACK` [12], is used only to *propose* endpoints; the interval is accepted only after two verified counts. For the k-th state the proposal is

$$a = \hat\lambda_{k-1} + \tfrac{1}{10}(\hat\lambda_k - \hat\lambda_{k-1}), \qquad b = \hat\lambda_{k+1} - \tfrac{1}{10}(\hat\lambda_{k+1} - \hat\lambda_k), \tag{7}$$

with the obvious modifications at the spectral edges. If $a < \eta < b$ and $N(b) - N(a) = 1$, the interval is accepted and $k = N(a)$ is reported as the *certified state index*. Otherwise a bisection fallback brackets the k-th

eigenvalue using only counts, and the resulting interval is verified in the same way. A proposal is never trusted: a corrupted or missing approximate spectrum can only make the construction fail, not make it wrong.

The two verified counts give more than isolation. They certify $\gamma := \min(\eta - a,\, b - \eta)$ as a lower bound on the distance from η to the rest of the spectrum, which is the quantity needed by both the Temple and the Davis–Kahan bounds.

2.4. The two bounds

Theorem 1 (Kato–Temple; classical [13, 14, 8])**.** *Let (a, b) contain exactly one eigenvalue λ of H_h, let $a < \eta < b$, and suppose*

$$\rho^2 < (b - \eta)(\eta - a). \tag{8}$$

Then

$$\eta - \frac{\rho^2}{b - \eta} \;\le\; \lambda \;\le\; \eta + \frac{\rho^2}{\eta - a}. \tag{9}$$

Note the crossed denominators: the lower bound uses the distance to the *upper* neighbour and the upper bound the distance to the *lower* one. Swapping them produces bounds that are numerically plausible and wrong; this was a real defect in the first version of our code and is now covered by a regression test. Condition (8) is also checked at run time: when it fails the certificate is marked invalid and says so, and it carries no enclosure of the requested state at all. The Weinstein interval is returned alongside as a diagnostic, not as a substitute: by Lemma 1 it says only that *some* eigenvalue lies within ρ of η.

Theorem 2 (Davis–Kahan [9])**.** *Under the isolation hypothesis of Theorem 1, let φ be the unit eigenfunction of λ. Then*

$$\sin \angle(\psi, \varphi) \;\le\; \frac{\rho}{\gamma}, \qquad \gamma = \min(\eta - a,\, b - \eta). \tag{10}$$

The angle bound requires only the verified gap, not condition (8). It is also the reason why the protocol reports the state index: if ψ happens to approximate the second eigenfunction while the caller asked for the first, (10) remains true but describes a different φ.

2.5. Resolution floor and the refusal of degeneracy

Floating-point counting has a resolution limit. If two eigenvalues are closer than the accuracy with which the inertia of $H_h - \sigma I$ can be computed, then $N(b) - N(a) = 1$ may be reported for an interval whose interior contains two eigenvalues, and every bound built on that interval is meaningless. We therefore refuse to certify when

$$\min(\eta - a,\, b - \eta) \le 10^{-9}\, s, \tag{11}$$

with

$$\begin{aligned} s &= \max\left\{1,\ \max_i |1/h^2 + V_i|,\ 1/h^2\right\} \quad \text{(one dimension)},\\ s &= \max\left\{1,\ 1/h_x^2 + 1/h_y^2 + \max|V|\right\} \quad \text{(two dimensions)}. \end{aligned} \tag{12}$$

The first entry is the largest diagonal entry of H_h or twice the off-diagonal one, and the two coincide when $V \ge 0$. The constant 10^{-9} is a heuristic safety margin of roughly 10^7 machine epsilons relative to the operator scale; it is not a theorem. It is chosen far above the observed noise of the count in our tests and far below any spectral gap of physical interest.

Remark 1 (Degeneracy is refused, not accommodated)**.** *If the state of interest is degenerate, no interval satisfying the isolation hypothesis exists, and the protocol returns* `valid=False` *together with the reason. This is correct behaviour, not a numerical failure. A residual of* 10^{-12} *for a degenerate eigenvector is possible, and a report that quotes only the residual will describe it as an accuracy of* 10^{-12}*; the certificate instead states that the state cannot be identified. The distinction matters for any automatic workflow that uses the certificate to select states, and it is demonstrated in Section 7.7. What the refusal does not mean is that nothing can be said at such a level: Section 4 replaces the isolated state by the invariant subspace, for which an isolating interval—one containing exactly the degenerate cluster—does exist and the same counting machinery verifies it.*

2.6. What is and is not certified

The enclosure (9) and the angle bound (10) are theorems about the exact matrix H_h; the implementation evaluates them in double precision. The arithmetic of the Rayleigh quotient and the residual is not the issue—it is a few $O(\varepsilon)$ perturbations of quantities that are themselves $O(1)$—but the count $N(\sigma)$ is a combinatorial object computed from a floating-point factorization, and a single misclassified sign changes it by one. The guards of Section 2.2 and

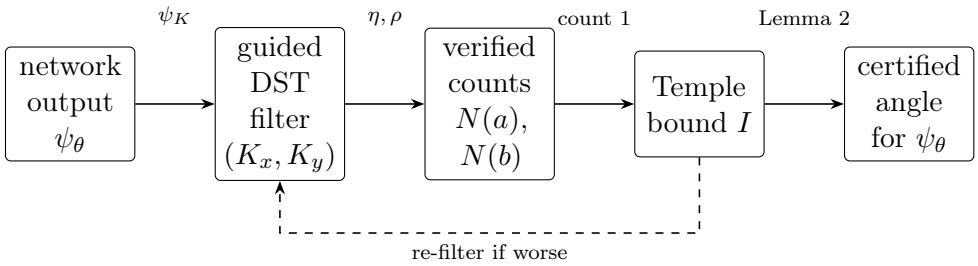


Figure 1: The certification pipeline. The counters are called for each candidate truncation pair; the filter keeps the narrowest enclosure among those whose verified state index agrees with that of the unfiltered output (Section 3.2). The parenthesised arrows are the feedback that makes the filter guided rather than fixed.

the floor (11) are designed so that the count is wrong only if the factorization is wrong by much more than its typical error, which we have not observed; but they do not constitute a proof. Section 6 closes this gap for the one-dimensional count and for two-dimensional separable potentials, by redoing the count in directed-rounding interval arithmetic at a factor of thirteen in cost; the general two-dimensional factorization remains floating point there, and we say why.

The second limitation is scope: everything above refers to H_h. A two-sided bound for the continuous operator would require an eigenvalue-perturbation estimate between H_h and $\mathcal{H}$ with explicit constants; Section 6.4 instead obtains a one-sided bound from the min–max principle, which needs no such estimate and costs one Rayleigh quotient. Section 7.8 measures this gap for the two potentials where the truncated-domain eigenvalue is known, and shows that at $N = 81$ the gap (3.5×10^{-4}) exceeds the certified width of a trained two-dimensional network (0.5–1.1×10^{-4}) by a factor of three to eight. The certificate is therefore already limited by the grid, not by the network: refining the network cannot buy accuracy that the discretization does not have.

3. The protocol

A certificate is only as meaningful as the object it is applied to. This section describes the three decisions that turn the machinery of Section 2 into a procedure that produces an honest number for a trained network. Figure 1 shows the pipeline; each subsection below justifies one arrow.

3.1. A residual whose energy cannot leak

The training objective is the squared residual of (3), a term that fixes the scale, a boundary term and an energy term,

$$\begin{aligned}\mathcal{L}(\theta) = \operatorname*{mean}_i \big(H_h\psi_\theta - \eta(\psi_\theta)\psi_\theta\big)_i^2 + w_n\big(\langle\psi_\theta,\psi_\theta\rangle_h - 1\big)^2 \\ + w_b\big(\psi_\theta(a)^2 + \psi_\theta(b)^2\big) + w_e\,\eta(\psi_\theta),\end{aligned} \tag{13}$$

with $(w_n, w_b, w_e) = (100, 100, 10)$ in one dimension. In two dimensions the residual and the energy are evaluated on the unit iterate $u = \psi_\theta/\|\psi_\theta\|_h$, the Dirichlet condition is imposed exactly by replacing the network output with $\psi_\theta \leftarrow M \odot \psi_\theta$, where $M(x, y) = (x-a)(b-x)(y-a)(b-y)$, so the boundary term vanishes identically and $w_e = 10$ is kept. In both dimensions $\eta(\psi_\theta)$ is the Rayleigh quotient of the iterate, treated as a constant in the backward pass (it is the minimizer over E, so detaching it leaves the gradient correct).

Four properties of (13) matter for certification. First, because the residual uses the same η as the certificate, the training objective and the certified quantity coincide; a network trained on a different energy (a learnable scalar, or a reference value) is optimizing a different residual, and the certificate then measures that other objective. Second, the scale must be fixed by an explicit term: without one $\psi \equiv 0$ is an exact stationary point of the residual, the loss is quadratic in the amplitude, and the amplitude is left wherever the initialization put it—in our unnormalized two-dimensional run it stayed as small as $\|\psi\|^2 = 6 \times 10^{-5}$ while the shape had already converged. One dimension fixes the scale with the penalty $(\|\psi\|_h^2 - 1)^2$; two dimensions divide by the norm inside the loss instead. Third, the boundary condition: in one dimension the soft penalty of weight 100 leaves endpoint values of order 10^{-4} and contributes a negligible part of the residual, whereas in two dimensions the same penalty is diluted by the $O(1/N)$ fraction of boundary nodes and leaves the residual at $O(1)$ (Appendix A records both failures), which is why the mask is used there. Fourth, the energy term $w_e\eta$ is variational: it pulls the iterate toward the lowest Rayleigh quotient and selects the ground state within the near-null space of the residual; because η is computed from the iterate, it carries no external value.

3.2. Certificate-guided spectral filtering

A trained network contains high-frequency components that contribute almost nothing to the Rayleigh quotient—which is stationary to first order—but

dominate the residual, which is not. Filtering them out turns an uncertifiable output into a certifiable one. Under Dirichlet conditions the natural basis is the tensor product of sine functions,

$$\psi(x_i, y_j) \approx \sum_{p=1}^{K_x} \sum_{q=1}^{K_y} c_{pq} \sin \frac{p\pi x_i}{L} \sin \frac{q\pi y_j}{L}, \tag{14}$$

with coefficients obtained by a two-dimensional type-I discrete sine transform (DST-I) [15], which is separable, orthogonal up to a diagonal scaling, and self-inverse.

The truncation order is chosen without any reference solution. For each candidate pair (K_x, K_y) on a prescribed schedule we filter, certify, and record $(\text{width}, \text{state index})$. The selected pair is the one with the smallest certified width *among those whose certified state index equals that of the unfiltered input*. The state-consistency constraint is not optional: without it the filter happily projects an excited state onto the ground-state sine, producing a much narrower certificate for the wrong function. On the runs below the filter changes ψ by a relative L^2 amount between 4×10^{-6} and 2×10^{-3}, and reduces the residual norm by a factor between 1.2 and 7×10^{9}—below threefold in half of the eighteen runs of the main sweep.

3.3. Transporting the bound to the raw output

The certificate of Section 3.2 refers to ψ_K, not to the network output ψ. The gap can be bounded by an elementary inequality, so that the network output itself receives a certified angle.

Lemma 2 (Transport)**.** *Let $u = \psi/\|\psi\|_h$, $u_K = \psi_K/\|\psi_K\|_h$, and let φ be a unit eigenfunction. Then*

$$\sin \angle(u, \varphi) \;\le\; 2\|u - u_K\|_h + \sin \angle(u_K, \varphi). \tag{15}$$

Proof. Since $\sin \angle(u, \varphi) = \|u - (u \cdot \varphi)\varphi\|_h$ and $|(u_K - u) \cdot \varphi| \le \|u - u_K\|_h$,

$$\begin{aligned} \sin \angle(u, \varphi) &\le \|u - u_K\|_h + \|u_K - (u_K \cdot \varphi)\varphi\|_h + |(u_K - u) \cdot \varphi| \\ &\le 2\|u - u_K\|_h + \sin \angle(u_K, \varphi). \end{aligned}$$

□

The first term is computed from the two saved functions; the second is Theorem 2 applied to ψ_K. The implementation uses the conservative factor 2 of (15); a factor 1 also follows from the Lipschitz property of sin on $[0, \pi/2]$.

The inequality is what allows the tables to report a bound for the network output even when the unfiltered output is not certifiable, and it answers the question "what exactly is being certified?" with a number rather than a caveat.

3.4. What a row of the results table contains

For each run the protocol reports: the certified state index and whether it matches the target; the energy η and the distance to an independently computed λ_h; the residual before and after filtering; the certified width before and after filtering; the certified angle bound, the transported bound for the raw output, and the true discrete angle (computed only for validation, never used by the protocol); the relative L^2 change caused by the filter; and whether the enclosure contains the reference. Reporting the raw and filtered columns side by side is essential: the first is what the network guarantees by itself, the second is what can be certified in its neighbourhood. Reporting only the second would overstate the network; reporting only the first would understate what the protocol can do.

The one case in which this row does not exist is the degenerate level: there is no single state to describe, and the certificate of Section 2 refuses. Its replacement—a record for the invariant subspace, with the multiplicity measured by counting—is the subject of Section 4, and it is the object that the experiments of Section 7.7 report.

4. Certified clusters: subspaces, not individual states

The certificate of Section 2 answers the question “which eigenvalue is this state?”. At a degenerate level that question has no answer: the problem does not define a single eigenvector, and a network that solves it exactly returns an arbitrary element of a two-dimensional space that depends on its seed. The honest object to certify is therefore the *invariant subspace*, and the honest statement is about the cluster of eigenvalues it spans. This section gives that statement; it is what turns the refusal of Section 1 into a positive result.

Guaranteed two-sided bounds for eigenvalue clusters, multiplicity included, already exist for conforming discretizations [16]. The setting here differs in one respect that changes what has to be computed: a network output is not a Galerkin projection, so no discretization parameter controls its error and no saturation hypothesis relates it to an enriched space. Both the cluster interval and the multiplicity must therefore be obtained from the count, and

the vector-side statement is a principal-angle bound that assumes nothing about how the trial subspace was produced.

4.1. The trial subspace and its residual

Let $\psi_1, \dots, \psi_m \in \mathbb{R}^{|I|}$ be m approximations—m network outputs, m snapshots of one training run, or m vectors from any other source—and let $X = [x_1, \dots, x_m]$ be an orthonormal basis of their span, $\langle x_i, x_j \rangle_h = \delta_{ij}$, obtained by symmetric orthogonalization $X = \Psi G^{-1/2}$, $G_{ij} = \langle \psi_i, \psi_j \rangle_h$. This choice matters: $G^{-1/2}$ is invariant under rotations of the input basis, so the certificate depends on $\operatorname{span}\{\psi_j\}$ and not on how the columns were ordered or mixed. If G is singular the m vectors do not span an m-dimensional space and the certificate refuses before computing anything else.

The compression of H_h to the trial subspace and its residual are

$$M = X^{\mathsf{T}} H_h X \in \mathbb{R}^{m \times m}, \qquad R = H_h X - XM, \qquad \rho_F = \|R\|_F, \tag{16}$$

and the Rayleigh–Ritz pairs (θ_i, y_i), $My_i = \theta_i y_i$ with $\theta_1 \le \cdots \le \theta_m$, give the Ritz vectors $x_i \leftarrow Xy_i$ (we reuse the symbol, since $\{Xy_i\}$ is again an orthonormal basis of the same subspace) and the residual

$$r_i = H_h x_i - \theta_i x_i = R y_i, \qquad \rho_i = \|r_i\|_h, \qquad \sum_i \rho_i^2 = \rho_F^2. \tag{17}$$

Everything in (16)–(17) is computed from the approximations alone.

4.2. Two propositions

Write $N(\sigma) = \#\{i : \lambda_i < \sigma\}$ for the verified count of Section 2.2, and suppose an interval (a, b) has been *verified* to contain exactly m eigenvalues,

$$N(b) - N(a) = m, \tag{18}$$

counting multiplicity. Let $V \subset \mathbb{R}^{|I|}$ be the corresponding invariant subspace, $\dim V = m$, and let P_V be the orthogonal projector onto it in $\langle \cdot, \cdot \rangle_h$. Define the cluster margin

$$\delta_i = \min(\theta_i - a,\, b - \theta_i), \qquad \delta = \min_i \delta_i. \tag{19}$$

Lemma 3 (Cluster localization)**.** *Assume* (18)*,* $\theta_i \in (a, b)$ *for all* i*, and* $\rho_i < \delta_i$*. Then*

$$\operatorname{dist}\big(\theta_i,\ \sigma(H_h) \cap (a, b)\big) \le \rho_i \qquad \textit{for every } i.$$

Proof. The pair (x_i, θ_i) is an approximate eigenpair in the sense of Definition 1 with $\|x_i\|_h = 1$ and residual r_i, so Lemma 1 gives $\operatorname{dist}(\theta_i, \sigma(H_h)) \le \rho_i$. Every eigenvalue of H_h outside (a, b) is at distance at least δ_i from θ_i, and $\rho_i < \delta_i$; hence the eigenvalue realizing the distance lies inside (a, b). □

Lemma 3 is the statement the single-state certificate cannot make: it is valid no matter how close the θ_i are to each other, because it never asks the interval to separate them. What it does not give is an identification: when $\rho_i \ge \delta_i/2$ the intervals $[\theta_i - \rho_i, \theta_i + \rho_i]$ may cover the same eigenvalue twice, and the certificate makes no claim about which cluster eigenvalue belongs to which Ritz value. At a degenerate level no such claim exists.

The angle between the trial subspace and V obeys a Davis–Kahan inequality in which the role of the spectral gap is played by the margin (19).

Theorem 3 (Principal angle)**.** *Under the hypotheses of Lemma 3,*

$$\sin\Theta\big(\operatorname{span} X,\, V\big) \;\le\; \frac{\rho_F}{\delta},$$

where Θ *is the largest principal angle between the two subspaces.*

Proof. Let $\{\varphi_j\}$ be an orthonormal eigenbasis of H_h with $H_h\varphi_j = \lambda_j\varphi_j$, and expand $x_i = \sum_j c_{ij}\varphi_j$. Since $H_h x_i - \theta_i x_i = \sum_j (\lambda_j - \theta_i) c_{ij}\varphi_j$,

$$\rho_i^2 = \sum_j (\lambda_j - \theta_i)^2 c_{ij}^2 \;\ge\; \delta_i^2 \sum_{j:\,\lambda_j \notin (a,b)} c_{ij}^2 \;=\; \delta_i^2 \left\|P_V^{\perp} x_i\right\|_h^2,$$

because an eigenvalue outside (a, b) differs from $\theta_i \in (a, b)$ by at least δ_i. The Ritz vectors form an orthonormal basis of $\operatorname{span} X$, so

$$\left\|P_V^{\perp} X\right\|_F^2 = \sum_i \left\|P_V^{\perp} x_i\right\|_h^2 \le \sum_i \frac{\rho_i^2}{\delta_i^2} \le \frac{\rho_F^2}{\delta^2},$$

and the sine of the largest principal angle is the spectral norm of $P_V^{\perp}$ restricted to $\operatorname{span} X$, which is bounded by the Frobenius norm. □

Theorem 3 is the quantitative statement that a set of network outputs spans the right subspace. For a single state ($m = 1$) it reduces to the Davis–Kahan bound (10) with δ in place of γ, which is the same inequality shifted by η; the cluster version is not a sharper theorem, it is the same theorem applied to the object that remains well defined when the eigenvalue is degenerate.

4.3. The cluster certificate

The certificate is assembled exactly as in Section 2.3, with m in place of 1:

1. *Rank.* Form G and refuse if it is numerically singular.
2. *Rayleigh–Ritz.* Compute M, θ_i, ρ_i, ρ_F from (16)–(17).
3. *Interval.* Propose a and b at the midpoints of the spectral gaps on either side of the block of m approximate eigenvalues that matches $\{\theta_i\}$ (ARPACK is used to propose, never to decide), then *verify* (18). If the proposal fails, bracket the two counting transitions by bisection and verify again; if that fails too, the refusal reason distinguishes a multiplicity mismatch from a subspace that spans more than one cluster.
4. *Accept* if δ exceeds the resolution floor (11) and $\rho_i < \delta_i$ for every i; then report the interval (a, b), the per-value enclosures $[\theta_i - \rho_i, \theta_i + \rho_i]$, and the angle bound of Theorem 3. Otherwise refuse, with the reason.

The cost is two verified counts—one sparse factorization each—one $m \times m$ symmetric diagonalization, and m applications of H_h; it does not grow with the number of requested eigenvalues beyond the m matrix–vector products, and it is negligible next to training.

Three properties are worth stating explicitly. First, m is *measured*, not asserted: the count decides whether the trial subspace has the dimension the invariant subspace has, and a subspace that missed a state is refused rather than certified with a smaller dimension. Second, Lemma 3 brackets each of the m eigenvalues to within ρ_i of a Ritz value, and the multiplicity itself is verified only down to the counting resolution (11); both hold independently of how accurate the network is: a poor subspace yields a poor angle bound but never a wrong multiplicity. Third, the interval may contain a block of adjacent distinct eigenvalues; degeneracy is the case that motivates the construction, not a special case of it.

The refusal reasons are meant to be actionable, and that suggests a procedural use of the certificate. A multiplicity mismatch says that the trial subspace missed a state of the cluster or covered more than one cluster, and tells the user how many vectors to supply or how to split them; $\rho_i \geq \delta_i$ says that the subspace is not an accurate enough approximation and that more training is required; a span whose second direction is the difference of nearly identical outputs says that the runs collapsed and that a further direction must be forced, which is done by deflating the accepted outputs as well and retraining. The loop *train, certify, deflate, retrain* is what produced the

accepted neural pair of Section 7.7, and its termination argument is the one familiar from deflated eigensolvers: each accepted direction is projected out of the residual that the next run minimises.

Remark 2 (What a cluster certificate does not say)**.** *It does not say that θ_i approximates λ_i in the matching order, and it does not give an $O(\rho^2)$ enclosure: for $m = 1$ the Kato–Temple bound of Theorem 1 is sharper and should be preferred whenever its hypothesis holds. It also does not resolve eigenvalues that the counting cannot separate, exactly as in* (11)*; what it does is to make the resolution floor the* only *obstacle, removing the structural one.*

Remark 3 (Do independent runs span a degenerate level?)**.** *It is tempting to certify a degenerate level by training m networks and certifying their span, and the answer depends on the data rather than on the construction. In our experiments three of four independent runs collapsed onto the same symmetry-broken representative (pairwise overlap > 0.9999) while the fourth found the orthogonal partner. The certificate accepted every pair containing the fourth run, with bounds of 5.6–6.3×10^{-3}; it refused the two remaining pairs of near-identical outputs, $(0, 1)$ and $(1, 2)$, whose second directions are differences of norm 2.4×10^{-3} and 6.7×10^{-3}—for both, the Ritz residual of that direction (*16.3 *and* 7.5*) exceeds the margin (*6.1 *and* 7.2*), so the span contains no invariant subspace of dimension two; and it accepted the pair $(0, 2)$, which omits the fourth run, only with a bound two orders of magnitude looser (*0.68*), which is the honest signal that its second direction was mostly noise. The certified object is the subspace, not the individual vector, and the certificate sorts sets of outputs by how much of a second direction they actually contain. When the answer is negative the second direction can be forced by deflating the first output as well, as in Section 7.7.*

5. Implementation

5.1. Structure

The package is organised in three layers. The `physics` layer (pure NumPy) provides analytic references and a Numerov integrator used to cross-check them. The `solver` layer contains the samplers, networks, losses and trainer, and—separately from training—the certification modules `certify` (1D), `filtering` (1D DST filter), `certify2d` (2D inertia counting and certificates), `filtering2d` (2D DST filter and transport) and `certify_subspace`

Call	Purpose
`certify_eigenpair(psi, x, V, isolate=True)`	1D certificate
`count_eigenvalues_below(x, V, sigma)`	1D Sturm count
`guided_filter(psi, x, V)`	1D filter + re-certification
`certify_eigenpair_2d(psi, grid, isolate=True)`	2D certificate
`count_eigenvalues_below_2d(grid, sigma)`	2D Sylvester count
`guided_filter_2d(psi, grid)`	2D filter + re-certification
`transported_angle_bound_2d(...)`	Lemma 2
`certify_subspace_1d(psis, x, V)`	1D cluster certificate
`certify_subspace_2d(psis, grid)`	2D cluster certificate
`certified_sturm_count_1d(x, V, sigma)`	count, directed rounding
`certified_separable_count_2d(grid, sigma)`	separable 2D count
`certified_eigenvalue_brackets_1d(...)`	certified eigenvalue brackets
`p1_rayleigh_bound(x, V, psi)`	upper bound on $\lambda_1(\mathcal{H})$
`q1_rayleigh_bound_2d(x, y, V, psi)`	the same in two dimensions
`principal_angle(X, exact, h)`	validation only, not a certificate

Table 1: Main entry points. A certificate is returned as a record with the energy, the enclosure, the verified gap, the state index, the angle bound and, when certification is refused, the reason.

(cluster and invariant-subspace certificates, Section 4), `certified_count` (directed-rounding counting and the certified eigenvalue brackets of Section 6) and `continuous_bound` (the variational bound of Section 6.4). The `scripts` directory contains one driver per experiment plus the aggregation and reporting utilities; every table in this paper is generated by a script from saved JSON, not transcribed by hand.

The certification modules do not import PyTorch and do not require a trained network: they operate on an array of samples and the grid. This is deliberate, because the certificate must remain meaningful if the training stack changes.

5.2. Application programming interface

Table 1 lists the entry points a user needs; everything else is built from them. The three certificate calls and the three counting calls are the whole interface to the numerics.

Every certificate is a plain record (a frozen dataclass), so it serialises to JSON without custom code; the experiment drivers store one record per run. A refused certificate carries a machine-readable reason—"no isolating interval exists", "gap below resolution floor", "Temple condition fails", "energy outside

the interval"—which is what makes the refusal auditable rather than a silent `None`.

A self-contained example, `examples/certify_demo.py`, exercises both paths in about four seconds and without PyTorch: it certifies the analytic trial function $e^{-x^2/2}$ of the harmonic oscillator, then reproduces the refusal-then-subspace sequence of Section 7.7 on a 61×61 box, where one member of the degenerate pair is refused as a state and the pair is certified as a cluster.

5.3. Drivers and reproducibility

The experiments of Section 7 are produced by the commands below.

```
python scripts/run_seed_sweep.py --jobs 6 --threads 4   # 1D, seeds 0-2
python scripts/run_seed_sweep_2d.py --jobs 2            # 2D, seeds 0-2
python scripts/filter_certificates.py                   # 1D certify
python scripts/filter_certificates2d.py                 # 2D certify
python scripts/exp7_degeneracy_2d.py                    # degeneracy
python scripts/exp8_subspace_certificate.py             # clusters
python scripts/exp9_resolution_boundary.py              # floor
python scripts/discretization_study_2d.py --sizes 41 61 81 101 121
python scripts/go_nogo.py ; python scripts/leakage_audit.py
python scripts/multistate_report.py
python scripts/bench_count.py                           # timings
python scripts/dense_reference_2d.py                    # 2D reference
python scripts/appendix_failures.py --variant soft_boundary
python scripts/residual_vs_error.py runs/appendix_failures/unnormalized
python scripts/lapack_reference_widths.py
python scripts/directed_count_report.py
python scripts/continuous_bound.py
python scripts/continuous_bound.py --alignment   # finite well, O(h) sign
python paper/make_tables.py ; python paper/make_figures.py
```

Each driver writes a JSON manifest with the wall-clock time of every job, so the runs are reproducible and the cost figures below are measured rather than estimated. The 1D runs (10 000 epochs, grid of 2001 points, network 64×3) take 30–45 s each on a laptop CPU; the 2D runs (6000 epochs, 81×81 grid, network 64×4) take 99–109 s each, manifest times that include start-up and saving. Certification itself is negligible in both cases. One count takes 0.4 ms at $N = 2001$ and 0.034 s at 81×81 (median of twelve timed calls, `scripts/bench_count.py`); a full certificate—interval proposal, verified count, and Temple bound—takes 0.074 s at $N = 2001$, which is some two hundred counts, and 0.19 s at 81×81, which is six, and a complete certificate-guided scan costs 2.6 s over the 27 one-dimensional cutoff orders; in two dimensions the timed benchmark records 16 s for an 81-candidate scan,

and the 30- and 62-candidate scans of the experiments, which examine fewer candidates, scale from it to about 6 and 12 s. The count inherits the scaling of the sparse factorization it reads: 3 ms at 41×41, 0.03 s at 81×81 and 0.85 s at 201×201, still small against a training run on the same grid.

5.4. Numerical safeguards and tests

The safeguards described in Section 2.2 are assertions in the code, not comments: a count that fails the identity-permutation or pivot test is retried with a perturbed shift and then reported as unavailable. The test suite (143 tests, run with or without `pytest`) locks the following properties, among others:

- the 1D Sturm count and the 2D Sylvester count agree element by element with dense eigen-decomposition on small grids;
- an exact discrete eigenvector receives a valid certificate whose enclosure contains the reference eigenvalue and whose width is at the round-off floor;
- a degenerate eigenvector is refused (Section 7.7), while the pair of exact eigenvectors spanning the level is certified as a cluster;
- a rank-deficient set of trial vectors, a single member of a degenerate pair, and a cluster whose multiplicity disagrees with the verified count are all refused, and a rotated basis of the same subspace returns the same certificate;
- the DST-I used by the filter is its own inverse;
- the guided filter recovers the exact sine mode of a box potential from a perturbed input;
- the transported bound of Lemma 2 dominates the true angle;
- the directed-rounding count agrees with dense eigen-decomposition on every shift where it does not refuse, is monotone in the shift, and *refuses* at a shift equal to an eigenvalue and for a non-separable two-dimensional potential;

- the variational bound exceeds the true continuous energy for the box and the harmonic oscillator, its error contracts at the observed $O(h^2)$ rate, its quadrature check rejects a kink it was not told about—and, in two dimensions, a non-polynomial potential—while accepting the quartic anharmonic one, and the two-dimensional Q1 bound equals the sum of the two one-dimensional bounds on a non-square grid.

The third item deserves emphasis: a test that only asserted "the certificate is valid" would have passed on a version of the code that produced a spuriously narrow interval for a degenerate state. The test asserts the refusal.

6. Certified counting and the continuous operator

Sections 2–4 certify quantities computed from H_h in double precision, and Section 2.6 names two limitations: the count $N(\sigma)$ is a combinatorial object read off a floating-point factorization, and every statement refers to H_h rather than to $\mathcal{H}$. This section attacks both with two constructions that turn out to be cheap: a directed-rounding count, which removes the first limitation in one dimension and in two whenever the potential is separable, and a variational bound on the continuous eigenvalue, which removes half of the second—the upper side—for every potential and needs no reference solution and no interpolation constant.

6.1. Directed rounding

An interval $[a, b]$ represents every real in it; an operation returns $\{x \circ y : x \in [a, b],\ y \in [c, d]\}$ rounded outward. The implementation is a few dozen lines: each endpoint is moved one ulp outward with `nextafter`, which is enough because IEEE 754 arithmetic is correctly rounded, so the exact result of a single operation lies within half an ulp of the computed one. This is not higher-precision arithmetic—the precision is still double—but every rounding is forced to the side that preserves the statement.

What the resulting count certifies must be stated carefully, because it is easy to overclaim. The certified matrix is the one the implementation actually stores: diagonal entries $h^{-2} + V_i$, off-diagonals $-\frac{1}{2}h^{-2}$, each taken as an exact real number. A certified count is therefore a theorem about *that* symmetric tridiagonal matrix. Any difference between it and the matrix a reader would write down from (1) is a modelling question (rounding of h^{-2}, of the potential samples), not a counting question.

6.2. One dimension: pivots rather than determinants

For a symmetric tridiagonal $T = H_h - \sigma I$ with diagonal a_k and off-diagonal b_k, the number of eigenvalues below σ is the number of negative pivots in the $LDL^\top$ factorization, i.e. of the recurrence

$$q_1 = a_1 - \sigma, \qquad q_k = (a_k - \sigma) - \frac{b_{k-1}^2}{q_{k-1}}, \tag{20}$$

by Sylvester's law of inertia. The determinant recurrence $d_k = (a_k - \sigma)d_{k-1} - b_{k-1}^2 d_{k-2}$ computes the same sign pattern in exact arithmetic, and we tried it first: it is unusable. In intervals the relative width of d_k exceeds 1—so the interval contains zero and the sign is undecidable—after 35 to 45 steps, on every one of the 363 scan shifts at $N = 2001$ (`runs/directed_count/stats.json`). The determinant cancels catastrophically because it alternates in sign and nearly does so at the scale of the spectrum's bulk. Equation (20) avoids this: when the pivot is resolved, the quotient b^2/q_{k-1} is a small correction to $a_k - \sigma$, and the interval stays narrow to the end of the recurrence. The cost is one division per node, which must be refused when the pivot interval straddles zero.

That refusal is the whole point. A count that cannot be decided is reported as uncertified with the index of the offending pivot, and the caller must handle it; it is never rounded to an integer. On three potentials at $N = 2001$, over 121 shifts spanning the low spectrum, 361 of 363 shifts were certified and every certified value equals both the floating-point count and the dense eigen-decomposition (`runs/directed_count/stats.json`). Both refusals are at the scan's last shift, which in each scan coincides exactly with an eigenvalue in double precision—precisely the case a rigorous implementation must refuse—and the refusal names the offending pivot (index 999). A median count costs 5 ms, against 0.4 ms for the floating-point version (`runs/_bench/count_timing.json`)—a factor of thirteen.

Two further uses follow. Bisecting on the certified count produces a *certified bracket* for a discrete eigenvalue; when the bisection reaches the resolution of the recurrence it stops and returns the last verified bracket instead of failing. This is what the driver `scripts/continuous_bound.py` reports in the λ_h column of Table 2 for the one-dimensional rows and, through the Kronecker route described below, for the two separable two-dimensional potentials as well; only the anharmonic potential's λ_h is a sparse-factorization eigenvalue rather than a certified bracket.

6.3. Two dimensions: a Kronecker sum, and why not elimination

The obvious two-dimensional analogue is a banded $LDL^\top$ in intervals. It does not work, and the failure is quantitative rather than a matter of implementation: the elimination divides by pivots and the interval width is amplified at each step, so pivots straddle zero long before the factorization ends. At 361 unknowns, 6 of 10 shifts could not be counted; the relative pivot width reached 0.93 immediately before one of the refusals, and the earliest refusal came at step 174 of the elimination, where the width was 0.44 (`runs/directed_count/stats.json`). Symmetric pivoting does not repair this, because the difficulty is overestimation, not a zero pivot.

For a separable potential, however, no elimination is needed at all. If $V(x, y) = V_x(x) + V_y(y)$ then

$$H_h = A_x \otimes I + I \otimes A_y \tag{21}$$

exactly, with A_x, A_y the one-dimensional operators on the two grids, so $\sigma(H_h) = \{\mu_i + \nu_j\}$ and

$$N(\sigma) = \#\{(i, j) : \mu_i + \nu_j < \sigma\}. \tag{22}$$

Each one-dimensional eigenvalue is bracketed by the certified count of the previous subsection, and the two-dimensional count is a comparison of sums. A pair whose sum interval straddles σ makes the count refuse. That is not a property of degenerate levels as such—any σ inside the certified bracket of a pair refuses—but in the cluster protocol the scan places σ on a level, and for the box, whose spectrum has multiplicities, the midpoint of a degenerate pair is the eigenvalue itself. On the harmonic and box potentials at 81×81, all 25 shifts of a scan were certified with zero disagreements against the floating-point inertia count, at 18 ms per count. The non-separable anharmonic potential is refused outright, with a reason, rather than silently falling back to floating point. Bisecting on this count yields certified brackets for the two-dimensional eigenvalues themselves—widths of 1.4–6.0×10^{-11} for the harmonic oscillator and 3.7×10^{-10}–2.5×10^{-9} for the box between 41 and 101 points per side—which is where the two-dimensional λ_h of Table 2 comes from.

6.4. A variational bound for the continuous operator

Every statement so far concerns H_h. The gap to $\mathcal{H}$ was left as an open item because the standard routes—an eigenvalue-perturbation estimate with

an explicit $O(h^2)$ constant, or a homotopy/Newton enclosure of the elliptic problem—need regularity or pointwise bounds that a trained network does not supply.

One direction needs none of that. By the min–max principle,

$$\lambda_1(\mathcal{H}) = \min_{u \in H_0^1} \frac{\frac{1}{2}\int u'^2 + \int V u^2}{\int u^2} \;\le\; R(u) \qquad \text{for every } u \in H_0^1, \tag{23}$$

so *any* conforming trial function yields an upper bound on the continuous ground-state energy. The P1 (bilinear, in 2D) interpolant of a sampled function is continuous and vanishes at the boundary, hence is in H_0^1: the network's own eigenfunction, once interpolated, is a legitimate trial function. No reference solution, no grid operator and no interpolation constant enters. This is the precise sense in which the certificate now reaches the continuous operator.

The arithmetic of (23) must be distinguished from its mathematics. In exact arithmetic the stiffness term is exact on our grids: piecewise constant in one dimension, and a polynomial of degree two in each variable in two dimensions, which the Gauss rule integrates exactly. The potential term is integrated by Gauss quadrature on each element, which is exact when Vu^2 is a polynomial of low degree there—true for the harmonic oscillator, the box, and, with the breakpoints supplied as element boundaries, the finite well. Rather than trust the hypothesis, the code evaluates the potential integral under two Gauss rules of different order and refuses the bound whenever the two intervals fail to intersect; a kinked potential whose kink lies inside an element is caught by that check. What the code does *not* do is evaluate the element integrals in interval arithmetic: the accumulations are outward-rounded, with a padding that grows with the number of terms, but the element-level arithmetic and the evaluation of V at the Gauss nodes are ordinary double precision. For the potentials here that rounding is a few ulp, orders of magnitude below the padding the summation already carries and eleven orders below the bracket it is compared with; it is nevertheless a gap, and a user who needs it closed must supply an interval evaluation of V. The one-sided statement—$\lambda_1(\mathcal{H}) \le R(\psi_h)$—is then a theorem about the computed quotient up to that rounding, and exactly a theorem about the quotient in exact arithmetic.

Table 2 applies (23) to the coarse and fine ends of each of the six problems. Two columns matter. The bracket $R(\psi_h) - \lambda_h$ contracts by the ratio that

Potential	N	λ_h	$R(\psi_h)$	$\lambda^\star$	in
Harmonic	501	0.4999819994	0.5000180013	0.5000000083	✓
Harmonic	4001	0.4999997318	0.5000002813	0.5000000083	✓
Infinite well	501	4.9347859593	4.9348184354	4.9348022005	✓
Infinite well	4001	4.9348016381	4.9348024542	4.9348022005	✓
Finite well	501	-48.9981281228	-48.9787820344	-48.9810479595	✓
Finite well	4001	-48.9773754207	-48.9809713144	-48.9810479595	×
Harmonic (2D)	41	0.9993927428	1.0021728615	1.0007820274	✓
Harmonic (2D)	101	1.0005601465	1.0010042177	1.0007820274	✓
Box (2D)	41	9.8645320539	9.8746788339	9.8696044011	✓
Box (2D)	101	9.8687926854	9.8704161708	9.8696044011	✓
Anharmonic (2D)	41	1.0962601793	1.1002100307	–	–
Anharmonic (2D)	101	1.0979179490	1.0985487577	–	–

Table 2: Discrete eigenvalue λ_h (the midpoint of a directed-rounding bracket wherever the potential is separable—all one-dimensional rows, and the harmonic and box in two dimensions; a sparse-factorization eigenvalue for the anharmonic potential) and the variational upper bound $R(\psi_h)$ of (23) on the P1/Q1 interpolant of the discrete eigenvector. The final column asks whether the reference value $\lambda^\star$ lies in $[\lambda_h, R(\psi_h)]$; "–" means no reference is known. The finite well's failure at $N = 4001$ is real—it fails already at $N = 2001$—and is discussed in the text.

$O(h^2)$ predicts when the grid is refined: 6.2–6.3 for the three two-dimensional potentials between 41 and 101 points per side (prediction 6.25) and 56–64 for the harmonic oscillator and the infinite well between 501 and 4001 (prediction 64; the finite well has no such ratio, because its bracket changes sign, as discussed below). At the coarse grid the reference value lies between the two columns in all five cases where it is known; at the fine grid in four of five, the exception being the finite well, discussed below. For the two-dimensional anharmonic potential, where no reference exists, the bracket is a *one-sided* statement: the true ground-state energy is at most the tabulated $R(\psi_h)$.

6.5. The two sides are not symmetric

The upper bound is unconditional; the lower side is not, and one of our own potentials shows why. The finite well has a discontinuous V, so the local truncation error of the three-point stencil is $O(h)$—not $O(h^2)$—near the jump, and its *sign depends on where the jump falls relative to the grid.* Across the twelve grids $N = 1000, \ldots, 1011$ on the same well, the discrete ground energy lies above the continuous value on six and below on six, with errors up to $1.4\times$

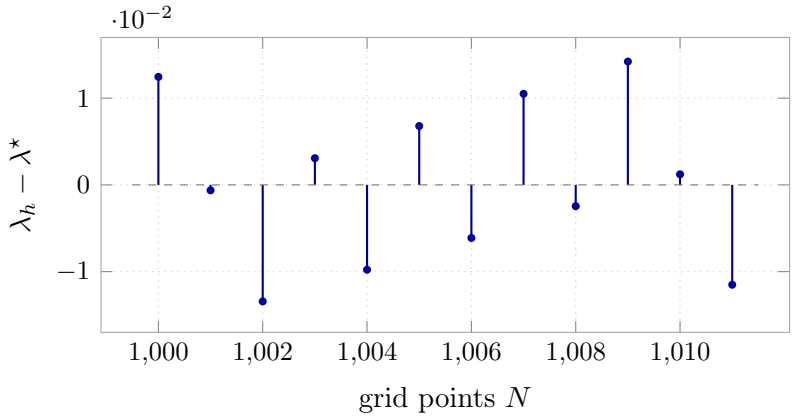


Figure 2: The finite well on twelve consecutive grids: the same potential, only the grid moved. The discrete ground energy alternates above and below the continuous value, with no trend in N, so the sign of the discretization error is a property of the alignment of the jump with the grid rather than of the grid size. Plotted from `runs/continuous_bound/alignment.json`; the $N = 4001$ point discussed in the text lies outside the window.

10^{-2} and no monotone decay (`runs/continuous_bound/alignment.json`). Figure 2 shows the twelve signs.

The mechanism is visible on a single grid. At $N = 4001$ the jump falls exactly on a node, and the value that node receives—inside the well or outside—is the whole error: keeping the grid and sampling that one point as -50 instead of 0 moves λ_h by 7.4×10^{-3}, from $+3.67 \times 10^{-3}$ to -3.73×10^{-3}, straddling the true value. The convention used in the table (the node outside, so the well is one grid point narrower) is what puts λ_h above $\lambda^\star$, and is why the finite-well bracket of Table 2 does not contain it at the finer grid. Neither convention is more accurate; the discrete energy of a discontinuous potential at an aligned grid is not a variationally consistent estimate of the continuous one, and no amount of rounding control changes that.

The lesson is not that the count is wrong—it is certified, and it is a certified statement about the stored matrix—but that a certified discrete calculation does not by itself certify a continuous quantity when the method is not variationally consistent. The same grid on a smooth potential behaves: for the harmonic oscillator the discrete value is below the continuous one at every grid we tried.

For the infinite well the lower side can also be made rigorous, and by an elementary inequality rather than a numerical observation. On $[0, 1]$ with

$h = 1/(n+1)$, the eigenvalues of $H_h = -\frac{1}{2}\Delta_h$ are

$$\lambda_j^{(h)} = \frac{2}{h^2}\sin^2\frac{j\pi h}{2}, \qquad \lambda_j = \frac{j^2\pi^2}{2}, \tag{24}$$

and $\sin\theta \le \theta$ gives $\lambda_j^{(h)} \le \lambda_j$ for every j and every h. In two dimensions the same argument applies along each axis, and since the ordering of sums is preserved, $\lambda_j^{(h)} \le \lambda_j$ holds for the entire spectrum of the two-dimensional box. There both ends are provable: the sine inequality puts the discrete eigenvalue below $\lambda^\star$, (23) puts $\lambda^\star$ below $R(\psi_h)$, and the certified bracket locates the discrete eigenvalue to 2.5×10^{-9}—five orders of magnitude inside the margin 8×10^{-4} between the two bounds at the finer grid, so the tabulated midpoint can be replaced by the bracket's lower endpoint without changing the conclusion. For the other potentials we do not have a lower bound, and we do not claim one: the honest summary is that (23) makes the continuous eigenvalue certifiable from above for every potential—for the harmonic oscillator the discrete value is below the continuous one at every grid we tried (`bound.json`)—and from below only where a comparison principle is available.

6.6. What this changes

The protocol of Section 3 is unchanged, and so is the cluster construction: what limits a certificate at a degenerate level is the resolution at which two shifts can be separated, and wherever the count is certified, directed rounding replaces "double precision with guards" by "provable or refused" at that same resolution. The experiments of Section 7 still count in floating point with the guards of Section 2.2, at 0.4 ms rather than 5 ms per count: the two agree on every shift at which the certified count does not refuse, and the certified function takes the same arguments when the guarantee rather than the speed is wanted. What does change is the answer to the question a referee should ask of any certified eigenvalue of a *discretized* problem: how far is this from the equation you actually wanted to solve? The upper bound answers it with a number, for every trial function the practitioner already has—including the trained one—and it is the only one of the two sides that holds without a comparison principle.

7. Numerical experiments

7.1. Setup

Potentials. In one dimension we use the harmonic oscillator $V = \frac{1}{2}x^2$ on $[-6, 6]$, the infinite square well on $[0, 1]$, and a finite square well of depth $V_0 = 50$ and half-width 1 on $[-8, 8]$. The first two have closed-form spectra, which we use only to construct training references and to label the runs; the certificate never sees them. In two dimensions we use $V = \frac{1}{2}(x^2 + y^2)$ on $[-3, 3]^2$, the box $[0, 1]^2$, and the non-separable potential $V = \frac{1}{2}(x^2+y^2)+\frac{1}{2}x^2y^2$ on $[-3, 3]^2$, for which no closed form is used at all: the certificate does not need one and neither does the reference, which is the dense LAPACK solve described under *References* below.

Networks and training. One-dimensional runs use an MLP of width 64 and depth 3 with Fourier features, 10^4 epochs of Adam, and a grid of 2001 points; two-dimensional runs use width 64, depth 4, no Fourier features, 6000 epochs and an 81×81 grid. The loss is (13). Three seeds (0, 1, 2) are run for every potential; all reported ranges are over seeds.

References. Every enclosure is compared against a value obtained independently of both the network and the certificate: in 1D by LAPACK's symmetric tridiagonal solver, evaluated as the Rayleigh quotient of the returned vector (the raw LAPACK eigenvalue has absolute error of order $\varepsilon\|H_h\|$, from 7×10^{-12} on the finite-well grid to 2×10^{-9} on the box grid, at least as large as the narrowest widths being tested); in 2D by a dense LAPACK decomposition of the full 6241×6241 matrix. ARPACK is used in 2D only to propose interval endpoints; it agrees with the dense reference to all ten printed digits and is never the reference itself. The reference is used for validation only; it is never an input to the certificate.

7.2. One-dimensional ground states

Table 3 summarises the nine ground-state runs with the leakage-free (Rayleigh-quotient) residual.

All nine enclosures contain λ_h, and after filtering all nine runs are certifiable. The raw certification rate is 7/9: two networks produce outputs whose residual is too large relative to the spectral gap for the Temple condition (8) to hold, which is exactly the situation in which an unfiltered claim would have to be withheld.

The three potentials separate sharply.

	Harmonic	Infinite well	Finite well
raw cert.	2/3	2/3	3/3
$\lvert E_f - \lambda_h \rvert$	1.6×10^{-6}–7.4×10^{-5}	3.7×10^{-14}–1.8×10^{-13}	$(1.3\text{–}7.7) \times 10^{-4}$
width, raw	0.143–0.65	$(5.9\text{–}6.2) \times 10^{-4}$	0.553–2.76
width, filtered	2.7×10^{-5}–9.3×10^{-4}	9.9×10^{-12}	0.374–0.729
angle bound	3.7×10^{-3}–2.2×10^{-2}	$(7.3\text{–}7.5) \times 10^{-10}$	0.56–0.835
contains λ_h	3/3	3/3	3/3

Table 3: One-dimensional ground states, three seeds per potential, 2001-point grid, 10^4 epochs. "raw cert." is the number of seeds for which the unfiltered network output is certifiable. Widths and angle bounds are from the filtered certificate; "contains" counts the enclosures that contain the independently computed λ_h.

- *Harmonic.* Certified widths 2.7×10^{-5} to 9.3×10^{-4}, angle bounds 3.7×10^{-3} to 2.2×10^{-2} rad against true angles of 9×10^{-4} to 5×10^{-3} rad. The enclosure is between 5 and 19 times wider than the true error—usable, and the only potential where it is.

- *Infinite well.* Width 9.9×10^{-12}, i.e. at the round-off floor of the interval endpoints. This is not evidence that the network is accurate to 10^{-11}; it is evidence that after filtering the network output is a single sine mode, and the residual of that mode is limited only by the arithmetic. We report it as a validation of the mechanism, not as a precision claim.

- *Finite well.* Widths 0.37–0.73 and angle bounds up to 0.84 rad, against a true energy error of $1.3\text{–}7.7 \times 10^{-4}$ and a true angle of 10^{-3}–3×10^{-3} rad. The enclosure is correct and useless: the ratio of certified width to true error is 10^3. The reason is structural—the kink at $|x| = 1$ makes the discrete eigenvalue converge at first order, and the network's short-wavelength error near the interface is indistinguishable, from the inside, from a genuine high-wavenumber component of the exact eigenfunction. No amount of training removes the ambiguity; only a different discretization (or a bound that uses regularity) would.

This spread is the main practical message of the paper: a certificate is not uniformly tight, and the ratio between certified width and true error is itself a quantity that must be reported.

Residual energy	raw	filt. contains	med. $\lvert\Delta E\rvert$	med. width	max width
Rayleigh (no leak)	7/9	9/9	4.9×10^{-6}	3.0×10^{-5}	0.729
Learnable scalar E_θ	5/9	9/9	7.2×10^{-6}	4.9×10^{-5}	2.72
Reference E (leaky)	8/9	9/9	6.4×10^{-6}	2.6×10^{-5}	0.628
Deflation (leaky)	0/3	3/3	0.109	0.218	0.486

Table 4: Leakage audit: same architecture, 10^4 epochs, only the energy in the residual changes. Median $\lvert\Delta E\rvert$ is measured against the discrete reference; widths are after filtering.

7.3. What the filter does

Figure 6(a) shows the certified width as a function of the truncation order K for one run per potential. The harmonic curve has the shape of a bias–variance trade-off: at $K = 8$ the truncation discards legitimate structure and the width is 0.31; the width falls by four orders of magnitude to 3.0×10^{-5} at $K = 15$, stays flat through $K = 20$, and then grows steadily as noise modes re-enter the residual, reaching 0.59 at $K = 1500$. The minimum is selected without any reference, from the certificate alone, and it coincides with the point beyond which the residual is dominated by components that do not affect the energy. For the infinite well the minimum is at $K = 1$ and the curve is monotone increasing: the exact state is the first sine mode and every additional mode is noise. For the finite well the minimum is broad and shallow ($K = 600$, width 0.37), again reflecting the inability of the basis to separate interface error from physics.

The relative L^2 change caused by the filter is 4×10^{-6}–1.6×10^{-3} over the eighteen runs of the main sweep (and at most 7×10^{-3} anywhere in the leakage audit), so the object being certified is not a different function in any meaningful sense; what is removed is a high-frequency tail that carries most of the residual and almost none of the energy.

7.4. Leakage audit

Table 4 compares four training variants that differ only in where the energy in the residual comes from. The Rayleigh-quotient variant (13) uses no external information. The learnable-scalar variant optimises E_θ jointly with the network. The reference variant uses the exact energy—a leak, since that is precisely the quantity being predicted. The deflation variant projects out analytic reference states to reach excited states—also a leak.

Three observations. First, the three variants that differ only in the energy in the residual are within a factor of two of each other on the smooth potentials,

so a single-potential comparison would have found nothing; the deflation variant trains a different state and is not comparable (median $|\Delta E| = 0.11$ against 10^{-5}). Second, on the finite well the reference variant is markedly better (0.34–0.63) than the learnable-scalar variant (1.87–2.72), because a learnable scalar does not converge within 10^4 epochs on a potential whose energy is -48.97 (it stalls near -9.7); the leak buys accuracy by supplying the very quantity that is hard to learn. Third, the Rayleigh variant recovers most of the leak's advantage without the leak (0.37–0.73), which is why it is the protocol's default.

A fourth observation concerns what an energy error hides. On the infinite-well grid, where an eigenfunction is a single sine mode, the learnable-scalar variant reaches $|\eta - \lambda_h| = 2.6 \times 10^{-7}$ in one seed—the Rayleigh quotient of the raw output, not of the filtered one—while the residual of that output is 0.53. The certificate accepts the run, the enclosure before filtering is 0.30 wide, 10^6 times the true error, and the angle bound is 0.56 rad: the energy is right to seven digits and the eigenfunction is located only to 32°. This is the case quoted in Section 1.

The honest conclusion of the audit is narrower than "the certificate detects leakage". The certificate describes the quality of a function; it cannot determine where a number came from. What it does provide is a fixed measurement scale: the audit becomes a controlled experiment, and the statement "method A is better than B" can be checked against a certified width rather than against a reference the method may have seen.

7.5. Multiple states

Reaching excited states requires an extra mechanism, because the residual functional is minimized by the ground state. We use deflation: after training the ground state with seed s, the network for state n is projected at every step onto the orthogonal complement of the previously trained states of the *same seed*—never onto analytic references. The projection is exact to machine precision ($|\langle \phi, \psi \rangle| \leq 2 \times 10^{-16}$ after re-projection).

Table 5 reports 27 runs, of which 25 contain the target state's discrete eigenvalue in the filtered enclosure. Both failures are in the harmonic chain at seed 2. The $n = 1$ run deflated from a correct ground state but converged to ψ_2 (certified state index 2 against a request for 1), and the $n = 2$ run inherited that wrong function: its deflation removed ψ_2, so it settled on ψ_1 (certified index 1). The certificate reported the mismatch in both cases, so the state-consistency check flagged the runs instead of silently reporting a valid

Potential	n	raw	match	contains n	med. $\lvert\Delta E\rvert$	med. width
Finite well	0	3/3	3/3	3/3	6.8×10^{-4}	0.652
Finite well	1	3/3	3/3	3/3	3.6×10^{-4}	0.426
Finite well	2	3/3	3/3	3/3	6.7×10^{-4}	0.523
Harmonic	0	2/3	3/3	3/3	4.9×10^{-6}	3.0×10^{-5}
Harmonic	1	2/3	2/3	2/3	3.7×10^{-6}	3.6×10^{-5}
Harmonic	2	3/3	2/3	2/3	1.2×10^{-6}	4.7×10^{-5}
Infinite well	0	2/3	3/3	3/3	7.1×10^{-14}	9.9×10^{-12}
Infinite well	1	3/3	3/3	3/3	4.2×10^{-10}	7.2×10^{-10}
Infinite well	2	3/3	3/3	3/3	1.2×10^{-9}	2.6×10^{-9}

Table 5: Multiple states with seed-consistent deflation. "match" counts runs whose certified state index equals the target; "contains n" counts runs that both match and whose enclosure contains the target state's discrete eigenvalue. Every valid enclosure in the chain contains *its own* reference eigenvalue; the two runs lost to the column are those certified to the neighbouring state.

	Harmonic (2D)	Box (2D)	Anharmonic (2D)
cutoff (K_x, K_y)	$(7, 7)$	$(1, 1)$	$(7, 7)$
raw cert.	3/3	3/3	3/3
$\lvert E_f - \lambda_h\rvert$	$(1.8\text{–}4.4) \times 10^{-6}$	$(2.5\text{–}2.8) \times 10^{-14}$	$(2.1\text{–}4.6) \times 10^{-5}$
width, raw	$(3.3\text{–}5.4) \times 10^{-4}$	$(1.4\text{–}2.9) \times 10^{-5}$	$(1.6\text{–}2.5) \times 10^{-3}$
width, filtered	4.6×10^{-5}–1.1×10^{-4}	2.0×10^{-11}	7.7×10^{-4}–1.4×10^{-3}
angle bound	$(5.2\text{–}8.0) \times 10^{-3}$	$(1.4\text{–}1.5) \times 10^{-13}$	$(2.0\text{–}2.7) \times 10^{-2}$
contains λ_h	3/3	3/3	3/3

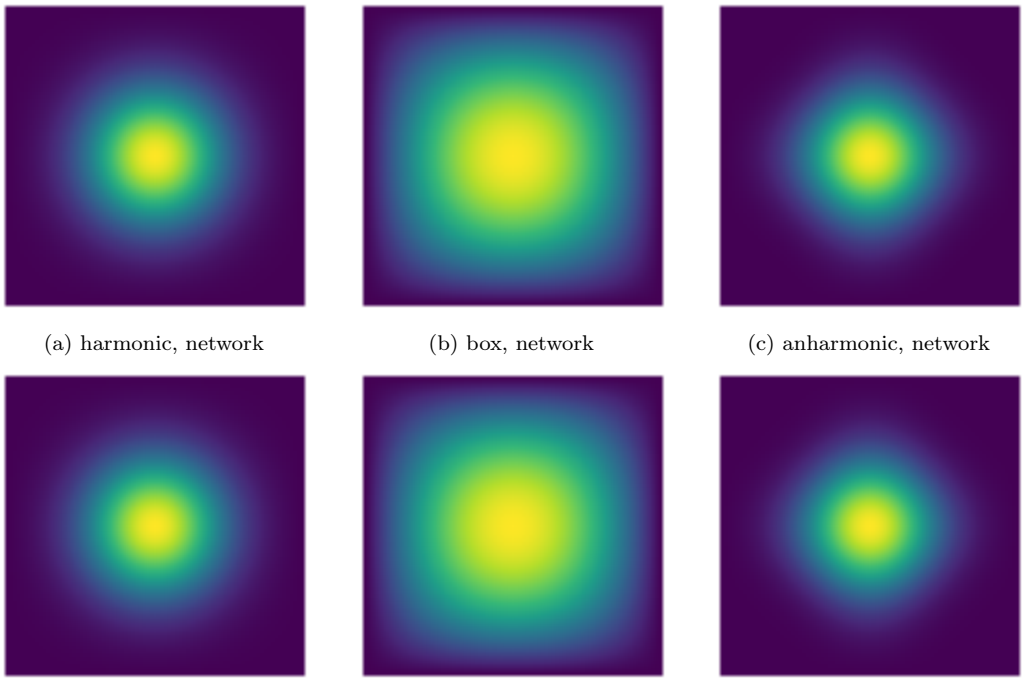
Table 6: Two-dimensional ground states, 81×81 grid, three seeds per potential. λ_h is computed by a dense LAPACK solve of the same matrix, independently of the network and of the ARPACK values that propose the intervals.

certificate for the wrong eigenfunction. This is the precise sense in which the state index is part of the certificate: a valid enclosure is a statement about *some* eigenvalue, not about the one the user wanted.

7.6. Two dimensions

The two-dimensional extension replaces the Sturm count by the Sylvester inertia of a no-pivot sparse LU (Section 2.2); the certificate logic is unchanged. Table 6 reports nine runs on an 81×81 grid.

All nine raw certificates are valid, all nine filtered certificates are valid, and all nine enclosures contain λ_h. For the box potential the filtered state is the exact discrete (1,1) sine mode, so the residual drops from 10^{-2} to



(a) harmonic, network (b) box, network (c) anharmonic, network

(d) harmonic, reference (e) box, reference (f) anharmonic, reference

Figure 3: Two-dimensional ground states: trained network (top) and discrete reference (bottom), each normalised to its own maximum. Every reference field is the ground eigenvector of the *same* H_h, returned by ARPACK on the same grid; its eigenvalue agrees with the dense LAPACK values of Table 6 to at least twelve digits. The reference is for display only and enters no error estimate. Colour maps are linear in $|\psi|$.

2×10^{-12} and the width to 2×10^{-11}—again a validation of the mechanism rather than a statement about the network. For the two smooth potentials the widths are 4.6×10^{-5}–1.1×10^{-4} (harmonic) and 7.7×10^{-4}–1.4×10^{-3} (anharmonic), with angle bounds that cover the true discrete angles in every run. The transported bound of Lemma 2 is valid in all nine runs; for the box it is 1.8–4.3×10^{-4} against a true angle of 0.9–2.1×10^{-4}.

Figure 3 shows the network output and the discrete reference for the three potentials. The visual agreement is what one expects from overlaps of at least 0.99997 where they are recorded; it is also exactly the kind of agreement that cannot be turned into an error statement without a certificate.

Input state	E	r	certified	width	outcome
Ground state	9.868336	3.3×10^{-12}	yes	2.0×10^{-11}	accepted
Excited (ARPACK 1)	24.663232	4.3×10^{-12}	no	–	refused
Excited (ARPACK 2)	24.663232	4.0×10^{-12}	no	–	refused
Excited (mixture)	24.663232	4.8×10^{-12}	no	–	refused

Table 7: Degeneracy experiment on the 81×81 box. The verified gap for the refused states is 7.1×10^{-9}, below the resolution floor 1.3×10^{-5} of (11).

7.7. Degenerate states are refused as states, certified as subspaces

Table 7 feeds *exact* ARPACK eigenvectors of the box potential into the single-state certificate. The ground state is certified with width 2.0×10^{-11}. The first excited level is doubly degenerate: the two eigenvalues differ by less than 2×10^{-13} (round-off; the exact value depends on the solver's start vector) and the eigenvectors are 4×10^{-12}-accurate, yet the certificate refuses both of them and a 0.43 mixture.

The refusal is the correct answer: no interval can contain exactly one of the two eigenvalues, so neither (9) nor (10) has a hypothesis, and the statement "the state is identified" is false even though the residual is 10^{-12}. The four rows make the point quantitatively: the residuals differ by less than a factor of 1.5 between the accepted ground state and the three refused excited states, so a convention that reports residuals alone would rank them as equally accurate, while the certificate's answer for the excited states is that no eigenfunction is defined and nothing is certified. In a workflow that ranks states or feeds them into a downstream calculation, that difference is decisive.

The same data, certified as a subspace. Table 8 applies the cluster certificate of Section 4 to the same eigenspace. Given the two exact eigenvectors the verified count returns $N(b) - N(a) = 2$ on the interval (17.2658, 32.0607), the Ritz residual is $\rho_F = 6.1 \times 10^{-12}$ at the margin $\delta = 7.397$, and Theorem 3 bounds the angle to the true invariant subspace by 8.2×10^{-13}, against a true angle of 2.3×10^{-16}. The eigenvectors that ARPACK returns for a degenerate level depend on its start vector, so the residual—and with it the last digit of the bound—moves between runs (8.2–8.5×10^{-13}); the verified interval and the acceptance do not, and it is those that the certificate asserts. The rotated basis $(x_1,\ 0.6x_1 + 0.8x_2)$ returns the same interval to all printed digits, which is the basis independence of the symmetric orthogonalization; the certificate depends on span and not on how the columns were chosen. The refusals are

Input	m	$N(b)-N(a)$	ρ_F	δ	$\Theta \le$	Θ_{true}	outcome
Exact pair	2	2	6.1×10^{-12}	7.4	8.2×10^{-13}	2.3×10^{-16}	accepted
Exact pair, rotated basis	2	2	6.3×10^{-12}	7.4	8.5×10^{-13}	3.0×10^{-16}	accepted
One member ($m = 1$)	1	–	4.4×10^{-12}	–	–	–	refused
Ground + 4th level	2	–	5.7×10^{-12}	–	–	–	refused
Networks 0, 1	2	–	16.3	6.08	–	–	refused
Networks 0, 2	2	2	4.6	7.31	0.68	2.8×10^{-2}	accepted
Networks 0, 3	2	2	4.1×10^{-2}	7.4	5.6×10^{-3}	3.0×10^{-4}	accepted
Networks 1, 2	2	–	7.53	7.18	–	–	refused
Networks 1, 3	2	2	4.5×10^{-2}	7.4	6.1×10^{-3}	2.4×10^{-4}	accepted
Networks 2, 3	2	2	4.7×10^{-2}	7.4	6.3×10^{-3}	2.5×10^{-4}	accepted
Seeds 0, 1, rotated	2	–	16.3	6.08	–	–	refused
Seeds 0, 3, rotated	2	2	4.1×10^{-2}	7.4	5.6×10^{-3}	3.0×10^{-4}	accepted
Network + partner	2	2	3.8×10^{-2}	7.4	5.2×10^{-3}	2.8×10^{-4}	accepted
Network + mix, partner	2	2	3.8×10^{-2}	7.4	5.2×10^{-3}	2.8×10^{-4}	accepted

Table 8: Cluster certificates on the 81×81 box. m is the number of vectors supplied, $N(b) - N(a)$ the *verified* multiplicity (a dash when the certificate refuses before the count is meaningful), ρ_F the Frobenius residual, δ the cluster margin (19), Θ the certified angle bound $\arcsin(\rho_F/\delta)$ and Θ_{true} the true principal angle, both in radians and the latter computed with a dense reference. In the neural block, runs 0, 1 and 2 converged to the same representative and run 3 to its partner; the certificate accepts the pairs that contain run 3, refuses $(0, 1)$ and $(1, 2)$, and accepts $(0, 2)$ with a much weaker bound. The last two rows use a direction forced by deflating the first output.

equally informative: one member of the pair alone ($m = 1$) is refused because no interval isolates it, and the set {ground, 4th level} is refused because any interval containing both of its Ritz values also contains the degenerate pair, so the count is 4, not 2. A subspace that skips a state is never reported as a smaller success.

Where the boundary lies: resolution, not degeneracy. Table 9 replaces the question "is this level degenerate?" by a measurement. Adding the anisotropy $\lambda(x^2-y^2)$ to the box splits the pair by $0.076\,\lambda$ with the mean energy unchanged to the printed digits, and the certificate is applied to both members at thirteen values of λ. The single-state certificate refuses every pair whose splitting lies below the counting floor—including splittings of 10^{-7} and 10^{-6}, six orders of magnitude above round-off and separated by ARPACK without difficulty—and accepts as soon as the splitting passes it: refused at split/floor = 0.95, accepted at 1.01, a factor of 1.06 between the two. The quantity compared with the floor is the verified gap, and its value is set by the interval construction. Above the transition the endpoints sit a tenth of the splitting inside each neighbour, so the gap is 0.9×split; at the crossing that proposal cannot clear

λ	split $\lvert E_1 - E_2\rvert$	floor	split/floor	single state	δ	$\Theta \leq$
0	$< 10^{-12}$	1.3×10^{-5}	< 0.01	refused	7.4	8.4×10^{-13}
1.0×10^{-5}	7.6×10^{-7}	1.3×10^{-5}	0.06	refused	7.4	7.5×10^{-13}
3.0×10^{-5}	2.3×10^{-6}	1.3×10^{-5}	0.18	refused	7.4	8.7×10^{-13}
1.0×10^{-4}	7.6×10^{-6}	1.3×10^{-5}	0.59	refused	7.4	8.1×10^{-13}
1.5×10^{-4}	1.1×10^{-5}	1.3×10^{-5}	0.89	refused	7.4	7.6×10^{-13}
1.6×10^{-4}	1.2×10^{-5}	1.3×10^{-5}	0.95	refused	7.4	7.6×10^{-13}
1.7×10^{-4}	1.3×10^{-5}	1.3×10^{-5}	1.01	accepted	7.4	8.1×10^{-13}
1.9×10^{-4}	1.4×10^{-5}	1.3×10^{-5}	1.13	accepted	7.4	7.6×10^{-13}
2.0×10^{-4}	1.5×10^{-5}	1.3×10^{-5}	1.19	accepted	7.4	7.4×10^{-13}
3.0×10^{-4}	2.3×10^{-5}	1.3×10^{-5}	1.78	accepted	7.4	7.7×10^{-13}
4.0×10^{-4}	3.0×10^{-5}	1.3×10^{-5}	2.37	accepted	7.4	9.8×10^{-13}
1.0×10^{-3}	7.6×10^{-5}	1.3×10^{-5}	5.94	accepted	7.4	8.3×10^{-13}
1.0×10^{-2}	7.6×10^{-4}	1.3×10^{-5}	59.37	accepted	7.4	8.2×10^{-13}

Table 9: Resolution boundary on the 81×81 box perturbed by $\lambda(x^2 - y^2)$, which splits the level linearly with the mean energy unchanged. "floor" is $10^{-9}s$ of (11). The single-state certificate accepts when the verified gap passes the floor; the gap is nine tenths of the splitting above the transition and equal to it at the crossing, so acceptance begins at split/floor ≈ 1. The cluster certificate has no such transition. Widths are omitted because the two-sided enclosure of an exact eigenvector is at the round-off floor (4.9×10^{-11}) in every accepted row.

the floor, the count-bisection fallback returns an endpoint at the neighbouring eigenvalue, and the gap is the splitting itself—which is why acceptance begins exactly at split/floor $= 1$ rather than at $1/0.9$. The cluster certificate, by contrast, accepts at every λ, exact degeneracy included, with the same margin $\delta = 7.397$ and a bound of 7.4–9.8×10^{-13}: its margin is the distance to the next level, not the internal splitting.

The practical content is that the phrase "degenerate state" is the wrong category for this protocol. What decides certifiability is whether two shifts can be separated in floating point, and a level split by 10^{-6} is as uncertifiable as one split by 10^{-13}. A practitioner who reports a pair of near-degenerate states separately, with a validated interval each, cannot support that claim on this grid; the pair as a subspace is certified to 8×10^{-13} in the same run.

Independent runs: which sets span the level?. Four additional runs, initialised differently and each deflated against the certified ground state, all reach the excited level: their energies agree with the discrete eigenvalue 24.663232 to 4.1×10^{-6}, their residuals lie in 2.7–3.4×10^{-2}, every one is refused by the single-state certificate (verified gap 3.8–4.1×10^{-6}, below the counting floor),

and every one lies in the exact eigenspace (fraction of its norm ≥ 0.99999996). They are nevertheless not four independent directions. Runs 0, 1 and 2 converged to the same representative—pairwise overlaps 0.99996–0.999997, mutual angles 0.14°–0.52°—while run 3 converged to the orthogonal partner, with overlap 0.01 with the others. Whether a set of runs spans the level is therefore an empirical question with no structural answer, and the certificate answers it for each set from the vectors alone:

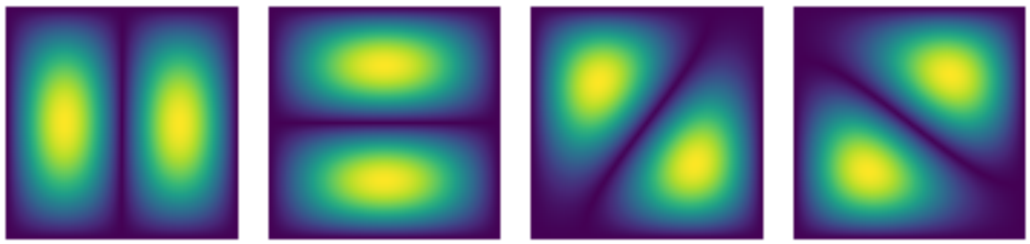
- $(0, 1)$ is refused. The two outputs differ by 2.4×10^{-3} in L^2, so the only second direction in their span is that difference, at grid scale; its Ritz residual is 16.25, far above the margin 6.08, and Lemma 3 has no hypothesis to stand on. $(1, 2)$ is refused for the same reason, at $\rho_F = 7.53$ and $\delta = 7.18$.

- $(0, 2)$ is accepted, but weakly: $\rho_F = 4.60$ gives the bound 0.68 against a true angle of 0.028. The certificate cannot know that this second direction is mostly noise; it can only report that the bound is two orders of magnitude looser than for a genuine pair—and it is still a valid bound.

- $(0, 3)$, $(1, 3)$ and $(2, 3)$ are accepted with $\rho_F = 4.1$–4.7×10^{-2}, $\delta = 7.397$ and bounds 5.6–6.3×10^{-3} against true angles of 2.4–3.0×10^{-4}.

No energy or residual separates these cases: the energies agree to 4×10^{-6} and the residuals lie in the same narrow band. Only the subspace certificate does, and it orders the evidence the way a reader would. Figure 4 shows the four functions behind the table: the two symmetry-related representatives that the runs converged to, next to the two exact modes they approximate.

A forced second direction. When the goal is to certify the level rather than to test whether training explored it, the second direction can be forced. Deflating the first network output in addition to the ground state—the same projection that is applied inside the loss—produces a function orthogonal to both (overlap 10^{-18} with the first output), which still lies in the level: 0.99999997 of its norm is in the exact eigenspace, and it coincides with the partner that run 3 found independently (overlap 0.99995). The pair is accepted with verified multiplicity 2, $\rho_F = 3.8 \times 10^{-2}$, $\delta = 7.397$ and an angle bound of 5.2×10^{-3} against a true principal angle of 2.8×10^{-4}. Rotating the basis ($0.6\psi_0 + 0.8\psi_\perp$ in place of $\psi_\perp$) leaves the interval, the multiplicity and the bound unchanged, as it must: the object certified is the subspace.



(a) network, seed 0 (b) network, seed 3 (c) exact, $\sin \pi x \sin 2\pi y$ (d) exact, $\sin 2\pi x \sin \pi y$

Figure 4: The doubly degenerate first excited level of the box, $|\psi|$ on the 81×81 grid, each panel normalised to its own maximum. The two network outputs are independent runs against the deflated ground state: one collapsed onto the representative found by three of the four runs, the other onto its orthogonal partner. They are visibly different functions—the nodal lines of one are the reflections of the other's—and they span the same eigenspace to within the certified angle.

The practical lesson is not that degenerate states are hard to train: all four runs reached the level to 4×10^{-6} in energy, and any of them would pass a residual-based test. It is that the quantities such a run reports—residual, energy, overlap with a chosen reference—are basis dependent at a degenerate level, and that whether a set of outputs spans the level is visible in none of them. The cluster certificate makes that question decidable: it accepts the spanning pairs with a bound, refuses the collapsed pair with a reason, and prices the difference between them in the bound itself. A degenerate level can therefore be certified without a reference solution, provided the dimension of the trial subspace is measured rather than assumed.

7.8. The discretization gap

Because the certificate refers to H_h, its width bounds the error against the discrete eigenvalue only; the distance to the continuous eigenvalue is a separate quantity, bounded from above by the variational principle of Section 6.4 and tabulated there. The two are independent: the box width below is 2×10^{-11} while the discretization error at the same grid is 1.3×10^{-3}. Table 10 quantifies that distance for the two potentials whose truncated-domain eigenvalue can be extrapolated, using verified count bisection at five grids; the extrapolated value λ^* agrees with the variational bracket of Table 2 at the corresponding grid, which is the check that the two independently computed descriptions of the same gap are consistent.

Two features deserve comment. The box potential extrapolates to $\lambda^* =$

Pot.	41	61	81	101	121	λ^*
Harmonic (2D)	1.4×10^{-3}	6.2×10^{-4}	3.5×10^{-4}	2.2×10^{-4}	1.5×10^{-4}	1.0007826
Box (2D)	5.1×10^{-3}	2.3×10^{-3}	1.3×10^{-3}	8.1×10^{-4}	5.6×10^{-4}	9.8696042

Table 10: Discretization error $|\lambda_h - \lambda^*|$ of the first eigenvalue of H_h; column headers are the grid size N and λ^* is fitted from $\lambda(h) = \lambda^* + ch^2$. The least-squares slope of $\log|\lambda_h - \lambda^*|$ against $\log h$ is 2.00 for both potentials.

9.8696042046, within 2.0×10^{-7} of π^2, with a clean second-order convergence; this is an end-to-end check of the counting, bisection and extrapolation chain. The harmonic potential is subtler: its infinite-domain eigenvalue is 1, but the Dirichlet problem on the truncated box $[-3, 3]^2$ has eigenvalue $\lambda^* = 1.0007826$, higher by 7.8×10^{-4} because the Gaussian tail is cut off. Consequently the raw error against the analytic value 1 *increases* with N (1.7×10^{-4} at $N = 61$ to 6.3×10^{-4} at $N = 121$) while the true discretization error decreases at second order. Reporting only the first column would look like a bug; reporting only the second would hide the truncation. Both are needed, and the certificate makes the distinction unavoidable.

At $N = 81$ the harmonic discretization error against λ^* is 3.5×10^{-4}, three to eight times larger than the certified width of the trained network (0.5–1.1×10^{-4}). On this grid the network is already better than the grid, and the certificate correctly reports the grid's limitation rather than the network's.

7.9. How tight is the enclosure?

Figure 5 plots the certified width against the true error for all 18 runs, together with the lines width $= 10^k \times$ error. Across the whole set the certified width lies between 5 and 3×10^3 times the true error, with a median ratio of 45 over all 18 runs (1.4×10^2 over the nine one-dimensional runs, 34 over the two-dimensional ones). The spread follows the potential rather than the dimension: 5–36 on the harmonic and anharmonic problems, 50–3×10^2 on the infinite well, and 7×10^2–3×10^3 on the finite well and the box. The two large groups are large for different reasons. On the infinite well and the box both the width and the error have reached the round-off floor (9.9×10^{-12} against 10^{-13} in one dimension, 2.0×10^{-11} against 2.5×10^{-14} in two), so the ratio is a ratio of round-off and says nothing about the method; on the finite well neither quantity is at round-off and the enclosure is genuinely wide, because the residual cannot distinguish a high-wavenumber error from a genuine component of the same shape (Section 8). For comparison, the

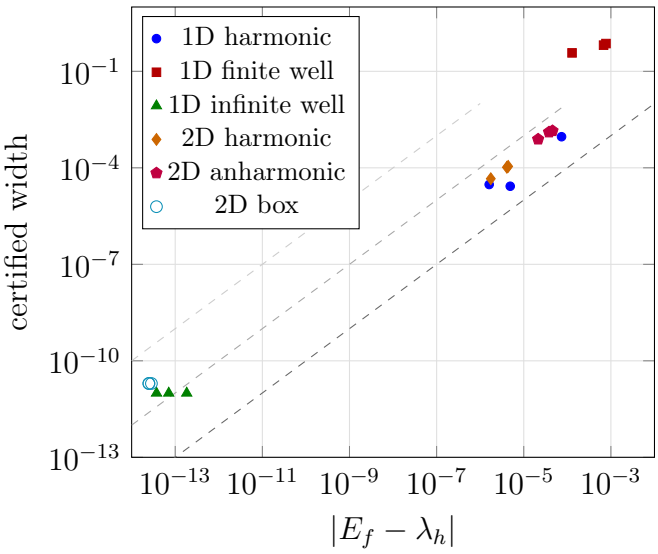


Figure 5: Certified width against true error. Dashed lines: width = error, 100× error, $10^4\times$ error. The points at the far left are the infinite-well and box runs, whose filtered states are exact sine modes; the upper-right outliers are the finite well.

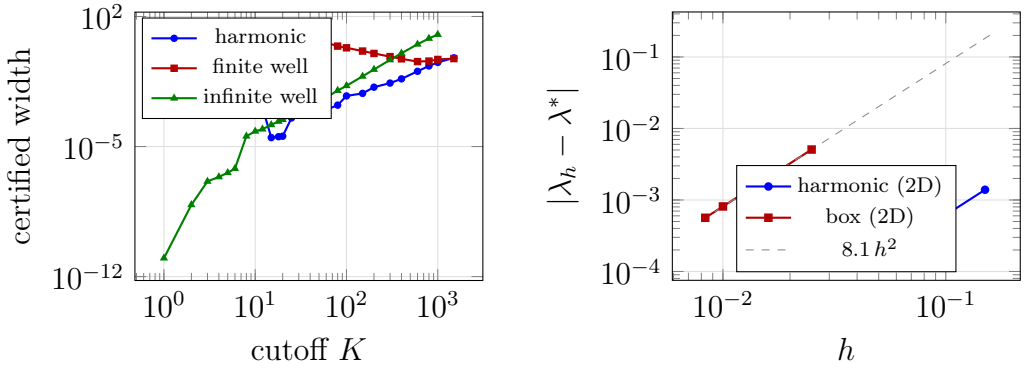
same certificate applied to a classical LAPACK eigenvector of the same H_h gives widths of 2.0×10^{-12} (harmonic), 9.9×10^{-12} (infinite well) and 9.8×10^{-11} (finite well), saved by `scripts/lapack_reference_widths.py` in `runs/reference1d/lapack_certificates.json`: the machinery itself is not the limitation, and a trained network's certified width is seven to ten orders of magnitude wider on the harmonic and finite-well potentials. The infinite well is the exception that proves the rule—after filtering, the network output is a single sine mode, so its certificate sits at the same round-off floor as the classical eigenvector's.

The figure also shows where the enclosure is genuinely informative: for the smooth harmonic and anharmonic potentials the ratio is 5–40, which is enough to certify that a network is accurate to, say, 10^{-4} without any reference. For the finite well the ratio is 10^3 and only the containment statement survives.

8. Discussion

8.1. What the protocol buys, and what it costs

The certificate turns an approximation into a statement with a hypothesis and a conclusion: *if* the interval contains exactly one eigenvalue (checked),



(a) width vs. truncation order

(b) discretization error vs h

Figure 6: (a) Certified width against the DST truncation order for one run per potential; the minimum is what the protocol selects without a reference. (b) Discretization error of the two-dimensional ground states against the $h \to 0$ limit; the dashed line is the expected second-order slope.

then the eigenvalue lies in $[\eta-\rho^2/(b-\eta),\ \eta+\rho^2/(\eta-a)]$ and the eigenfunction is within $\arcsin(\rho/\gamma)$ of ψ. The hypothesis is the part that classical presentations leave to the user; making it a computation is what allows the bound to be used on a mesh-free approximation whose spectrum is unknown.

The cost is small. In two dimensions a certificate requires four to six verified counts, each a sparse factorization taking 0.03 s at 81×81; in one dimension a count takes 0.4 ms, so the count-only bisection fallback can be used freely and a certificate spends some two hundred of them. Measured end to end, a full certificate takes 0.19 s and 0.074 s against 30–110 s of training. The guided filter repeats the certificate at each candidate cutoff—27 candidates in the one-dimensional runs and 30–62 in the two-dimensional ones—which costs 2.6 s and, scaled from the measured 16 s at 81 candidates, 6–12 s, still seconds. There is no reason not to certify every run that is reported.

The benefit is not accuracy but accountability. In our own experiments the certificate changed decisions four times: it rejected an excited-state run that had a valid-looking energy (state index mismatch), it exposed a degenerate state masquerading as a 10^{-12}-accurate one, it showed that part of what looked like network error on the finite well was discretization error that no amount of training could remove, and it decided which pairs of independently trained networks actually span a degenerate level—accepting the three pairs

that contain the orthogonal partner with bounds of 5–6 $\times 10^{-3}$, refusing the two collapsed pairs, and pricing the ambiguous pair at 0.68.

8.2. Limitations

The enclosure can be uninformative. For the finite well the ratio of certified width to true error is 10^3 and the angle bound is $O(1)$. The bound is correct and useless. This is not a defect of the implementation but a property of residual-based bounds on non-smooth problems: the residual cannot distinguish a genuine high-wavenumber component from an error of the same shape, and the spectral gap at the relevant scale is large enough that the quadratic Temple gain does not compensate. Any claim that such bounds "certify PINN accuracy" must be conditioned on the potential.

Degeneracy is now inside the protocol, but only at counting resolution. The cluster certificate of Section 4 removes the structural obstacle: a degenerate level is certified as an invariant subspace, with the multiplicity measured by the count and the angle bounded by Theorem 3. Two limitations remain, and they are of different kinds. The first is the resolution floor (11): two shifts closer than $10^{-9}s$ cannot be separated by the count, and a cluster whose boundary margin falls below the floor is refused—including, as the degeneracy experiment shows, a pair that double precision can barely split. The floor is a declared safety margin rather than a theorem (Section 2.5), and directed rounding does not remove it: it makes the count beneath the margin exact, so that the refusal happens for a stated reason at a stated pivot, rather than moving the margin. Counting at a higher precision would move the floor but not abolish the need for one. The second limitation is structural: a trial subspace that omits a member of the level is refused because its dimension disagrees with the count, which is the correct answer but means that the certificate can detect a wrong basis and cannot repair it. A milder limitation is that the cluster bound is first order in ρ: unlike the Kato–Temple bound it does not improve quadratically as the residual shrinks, because a cluster has no interior eigenvalue to isolate. For very accurate subspaces the counting resolution, not the angle bound, becomes the limiting quantity.

The certified object is discrete. The gap between H_h and $\mathcal{H}$ is $O(h^2)$ for smooth eigenfunctions and dominates the certificate at the grid sizes used here (Section 7.8). A certificate for the continuous operator would require either an explicit $O(h^2)$ bound with computable constants, or a homotopy-based enclosure of the elliptic problem itself [17], which needs a priori pointwise

bounds on the eigenfunction that a trained network does not supply; nor is a p-version/spectral discretization with exponential convergence implemented. We do not claim any of the three. What Section 6.4 does claim is the one-sided half, and it is enough to change what a reported energy means: the interpolant of the network eigenfunction is a conforming trial function, so its Rayleigh quotient bounds the continuous eigenvalue from above without any regularity input. The other side is where the difficulty sits, and the finite well shows it is not a technicality: a certified discrete eigenvalue can lie *above* the continuous one when the potential is discontinuous, so no care with the factorization turns a discrete count into a continuous enclosure. The same boundary explains why the count refers to H_h and not to the spectrum of a non-conforming trial space: such a space can carry spurious eigenvalues [18], so a discrete eigenvalue computed from it would not by itself identify the state it belongs to.

Directed rounding in one dimension, not in general two. The count of Section 6 is computed with outward rounding, so in one dimension—and in two whenever the potential is separable, which covers the harmonic oscillator and the box—it is a theorem about the stored matrix rather than a guarded estimate. What remains outside is the general two-dimensional count for a non-separable potential: the interval elimination refuses on most shifts at the grid sizes we use, so the anharmonic experiments still read their count off the floating-point factorization with the checks of Section 2.2. We report that measurement rather than route around it. The separable construction suggests the next step—splitting a non-separable potential into separable pieces and carrying the remainder as a perturbation—but we have not implemented it, and we do not claim it here.

Statistical power. We report counts, not rates with confidence intervals. Every enclosure we report contains the discrete eigenvalue to which it refers—the eighteen of the main sweep, the nine learnable-energy and nine reference-energy runs of the leakage audit, and the three deflated runs, for 39/39; in the multi-state chain each enclosure contains the eigenvalue of the state that its certified index names. The Wilson 95% interval for 39/39 is $[0.91, 1.0]$, so the data are still consistent with a per-run failure probability as large as 9%; the eighteen runs of the main sweep alone give $[0.82, 1.0]$. The multi-state chain is the more demanding test and is not perfect: 25/27 certify the *requested* state, Wilson $[0.77, 0.98]$, with both failures traced to an upstream state

mismatch that the state index flagged. A claim of the form "the method certifies correctly with probability at least 0.95" would require about seventy runs with no failure, and we do not make it.

8.3. Why not solve H_h instead?

A referee may reasonably ask what the certificate adds, since on every problem in this paper H_h is assembled and a shift-invert Lanczos solve of the same matrix returns λ_h and its eigenvector in 20 ms at 81×81 and 1.7 ms at $N = 2001$, roughly ten times faster than the certificate in two dimensions and forty times in one (0.19 s and 0.074 s; both measured in `scripts/bench_count.py`). We do not think the answer is speed, and the protocol should not be sold on it. Three things are at stake that a solve of H_h does not supply.

The certificate does not need a converged eigenvector. Counting is the only operation it performs on H_h, and ARPACK is used to propose intervals that a count then verifies; the count-only fallback of Section 2.3 removes even that. The statement therefore attaches to the object the user has—one network output, one snapshot, one subspace—rather than to the object a solver would produce instead.

At a degenerate level the classical comparison is ill-posed. The eigenvector a solver returns for a multiple eigenvalue is an arbitrary element of the level, chosen by its tie-breaking; the overlap of the same network output with it can be 1 or 0 without anything about the network changing. The experiments of Section 7.7 are exactly this case: three runs and the solver's eigenvector coincide, the fourth does not, and every residual-based comparison is silent. The cluster certificate is invariant under the choice, and it prices the difference. This is the one place where the certificate states something that *cannot* be read off a solver's output.

The audit is of the procedure, not of a number. State index, two-sided enclosure, angle bound and measured multiplicity together describe a pipeline end to end; a comparison with a reference eigenvalue describes one run. The leakage audit (Section 7.4) and the state-index failures of the multi-state chain (Section 7.5) are decisions of the first kind. This is also the honest limit of the present work: when H_h cannot be assembled—the high-dimensional regime that motivates neural eigen-solvers in the first place—every statement here refers to a matrix that does not exist, and matrix-free verification is open. The protocol is addressed to the intermediate regime in which the problem is

small enough to assemble and large enough that the learned function, not the solver, is what is being reported.

8.4. Relation to existing work

Enclosure of eigenvalues by variational bounds plus a verified count is the standard toolkit of a posteriori eigenvalue analysis for finite elements [7]: the Temple and Lehmann–Goerisch bounds are used exactly as here, and the count is usually obtained from a Cholesky factorization of the stiffness matrix at a shift. Our contribution is not to that theory but to its transfer: the approximation is a network, the matrix is assembled for the certificate alone, and the two failure modes that matter in this setting (degeneracy and leakage) are given explicit treatments.

For clusters the closest antecedent is the work of Cancès et al. [16], who give guaranteed two-sided bounds for eigenvalue clusters of conforming discretizations, multiplicities included. Their analysis is built on the Galerkin structure—residual estimators, orthogonality, and a comparison between the approximation space and its neighbourhood. A trained network has none of these properties, and applying those bounds to one would require proving an analogue from scratch. The counting route taken here needs no such hypothesis: it measures the multiplicity directly and bounds the subspace by the residual-to-margin ratio, which is why the same statement covers exact, approximate and accidentally degenerate clusters.

For forward PDEs, a posteriori error estimates for physics-informed networks have begun to appear [19]; they bound the error of a solution operator in a norm, and require regularity assumptions on the data. An eigenvalue problem differs in that the object to be bounded is a spectral quantity and the unknown is defined only up to a subspace; the counting machinery is what replaces the coercivity argument.

Verified-computing libraries such as INTLAB [20] and Arb [21], and their Python interfaces, would supply the same arithmetic at arbitrary precision. We use only IEEE doubles with one-ulp outward padding (`nextafter`) because the recurrences need nothing more: the certified count is a sign test on pivots whose intervals close under a correctly rounded operation, and the failure mode—a pivot interval that contains zero—is detected and refused rather than hidden. The same libraries remain the natural route to a fully rigorous two-dimensional count and to interval enclosures of the Rayleigh quotients themselves, at a modest constant-factor cost.

8.5. A reporting checklist

The experiments suggest a short checklist for papers that report neural eigenpairs, each item of which would have changed the interpretation of at least one run in this study:

1. state the discrete operator and the grid; report the discretization gap separately from the network error;
2. report the certified state index, not only the energy;
3. report the enclosure width, and the ratio of width to true error, not only the true error;
4. report raw and filtered quantities side by side, with the L^2 change caused by the filter;
5. state whether any reference quantity entered the training objective;
6. refuse to certify a degenerate state individually rather than reporting its residual as an accuracy, and certify the invariant subspace instead, with the multiplicity measured by counting.

Items 2, 3 and 6 are cheap; they require no additional theory and would make the reported numbers comparable across papers.

9. Conclusion

We have described a protocol that makes classical a posteriori eigenvalue bounds computable for neural eigenpairs of the discrete Schrödinger operator, and an implementation of it that runs on ordinary hardware. The essential element is not any of the inequalities—they are textbook—but the verification of their hypothesis: by counting the spectrum below two shifts with Sturm sequences or Sylvester inertia, the interval that the theorems require is constructed and checked rather than assumed. The same machinery supplies the spectral gap that turns a residual into a two-sided eigenvalue enclosure and an eigenfunction angle bound. At a degenerate level the same machinery is applied to the invariant subspace rather than to a state: the multiplicity is measured by the count, the cluster receives a two-sided enclosure, and the angle between the trial subspace and the true one is bounded, so that the structural obstacle of degeneracy becomes a certificate rather than a refusal.

The experiments on six potentials in one and two dimensions give a concrete picture of what can and cannot be certified. All 39 filtered enclosures we report contain the discrete eigenvalue they refer to—18 from the main

sweep and 21 from the leakage audit, the two-dimensional ones checked against a dense LAPACK solve rather than the ARPACK values that propose the intervals—and the certified width is between 5 and 40 times the true error on the harmonic and anharmonic potentials, between 50 and 3×10^2 times on the infinite well and 7–8×10^2 on the box—where width and error are both at round-off—and up to 3×10^3 times on the finite well, where the bound is correct but uninformative. A degenerate state with a 4×10^{-12} residual is refused as a state while the same eigenspace is accepted as a two-dimensional invariant subspace, with a measured multiplicity of two and a principal-angle bound of 8×10^{-13} on exact inputs and 6×10^{-3} on network inputs. Of four independently trained networks on that level, three collapsed onto the same representative and one found its partner; the certificate refused the pairs that do not span the level and accepted the pairs that do. Splitting the level by a controlled anisotropy locates the boundary between the two certificates exactly at the counting resolution—a split of 10^{-6} is refused as a state just as a split of 10^{-13} is, while the subspace is certified in both cases—so the obstacle is computable rather than structural. The network error on the 81×81 grid is smaller than the discretization gap, so the certificate correctly reports the grid as the limiting factor.

The implementation, the drivers that regenerate every table, and the test suite that locks the guards are available with this article. The two open items of the previous version of this work are closed here in part. The count is now computed in directed-rounding interval arithmetic—provably in one dimension, and in two dimensions whenever the potential is separable, which covers two of our three two-dimensional problems; the general sparse factorization still refuses too often in intervals to be useful, and we report the measurement rather than hide it. And the continuous operator now receives an upper bound from the variational principle—exact in exact arithmetic, and limited only by the floating-point evaluation of the quotient—at the price of a Rayleigh quotient rather than of an interpolation constant; the lower side exists only where a comparison principle does, which we demonstrate for the infinite well and the box and show to fail for the finite well. Neither addition changes the protocol, and neither touches the cluster construction: what limits the certificate at a degenerate level is the resolution at which two shifts can be separated, not the requirement that the level be simple.

CRediT authorship contribution statement

Chunyang Chen: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Writing – original draft. **Yuhao Peng:** Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Writing – review and editing, Supervision. Both authors contributed equally to this work and approved the final manuscript.

Code availability

The implementation, the experiment drivers and the scripts that regenerate every table and figure of this paper are part of the repository accompanying this article, `https://github.com/YuhaoPeng1103/PINN_equation` (MIT licence). A frozen archive will be deposited in the CPC Program Library.

Data availability

All data supporting this study are contained in the repository cited above: the measured artefacts behind every table and figure are stored under `runs/` as JSON, CSV and NumPy archives, and the drivers that regenerate them are in `scripts/` and `paper/`. No other data were generated or analysed.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude (Anthropic) in order to assist with drafting and editing the manuscript text and with the implementation and testing of the software. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Appendix A. Implementation notes

Appendix A.1. Training failures that shaped the implementation

Three failures were reproducible and are worth recording, because each one produces a network that looks superficially fine but cannot be certified. The runs behind them are kept in `runs/appendix_fourier2d/` and `runs/appendix_failures/`; the Fourier and hard-mask runs are outputs of `scripts/exp6_certificate_training_2d.py`, the soft-boundary and unnormalized runs of `scripts/appendix_failures.py`.

Random Fourier features in two dimensions. The one-dimensional runs use random Fourier features [22] and profit from them. In two dimensions the same configuration (scale 2.5) stalls: at 6000 epochs, on the harmonic potential whose discrete eigenvalue is 1.0004, it reaches $E = 1.5624$ and overlap 0.877 (the 1500-epoch snapshot is $E = 1.5636$ and 0.877) against $E = 1.0004$ and overlap 0.99997 for the plain MLP, and the single-state certificate refuses the output with a residual of 1.17 (`runs/appendix_fourier2d/`). Each feature $\sin / \cos(\omega_1 x + \omega_2 y)$ must cover a direction in the plane, and the sample efficiency of a fixed random direction set degrades with dimension. The 2D driver therefore disables Fourier features by default. This is an empirical configuration choice, not a claim about Fourier features in general.

The zero function in an unnormalized loss. If the loss is the residual of ψ itself rather than of the unit vector $\psi/\|\psi\|$, the zero function is a global minimum and not merely a stationary point: the Rayleigh quotient is scale invariant, so the residual is exactly linear in ψ and $L(c\psi) = c^2 L(\psi)$, whence $L(0) = 0 \leq L(\psi)$. No term then fixes the scale: the amplitude stays wherever the initialization put it, and any tolerance can be met by shrinking rather than by improving the shape. In a 3000-epoch run at $N = 41$ the field ended at $\|\psi\|^2 = 6.0 \times 10^{-5}$, a maximum amplitude of 1.5×10^{-2} (rms 7.5×10^{-3} over the grid), with a shape that is already accurate (overlap 1.0000 after normalization, energy error 1.1×10^{-5}). Measured on the field as returned, without normalization, the residual norm is 2.9×10^{-4} while the L^2 distance from the discrete ground state is 0.99—a residual three thousand times smaller than the error it is supposed to measure (`scripts/residual_vs_error.py`). Every diagnostic computed on the normalized field therefore looks excellent while the field itself carries a scale that the loss does not determine, and the training loss then measures the amplitude rather than the error. Fixing the scale by an explicit term—the penalty $(\|\psi\|_h^2 - 1)^2$ in one dimension, division

by the norm in two (equation (13))—removes the scale direction and excludes the zero solution; the denominator is clamped at 10^{-30} to keep the gradient finite during the first steps.

Soft boundary penalties in two dimensions. A boundary penalty with weight 100 does not enforce a Dirichlet condition to anything like machine precision, because the constrained nodes are an $O(1/N)$ fraction of the grid while the five-point Laplacian amplifies their error by h^{-2}. After 3000 epochs at $N = 41$ on the box potential the penalty leaves the residual at 5.72 and the energy error at 5.0×10^{-2} ($E = 9.9144$ against the discrete 9.8645), with boundary nodes carrying values up to 0.22. The certificate still returns `valid`—its hypotheses hold for whatever field it is given—but the enclosure is 4.68 wide against a true error of 5.0×10^{-2}, more than five orders of magnitude looser than on a hard-constrained field. Multiplying the network output by the mask $M = (x - a)(b - x)(y - a)(y - b)$ enforces the boundary condition exactly and is a statement of the problem, not of the solution; with everything else unchanged the same configuration reaches residual 1.2×10^{-2}, energy error 1.2×10^{-6}, enclosure width 2.0×10^{-5} and angle bound 9.0×10^{-4} (`runs/appendix_failures/`). The mask costs expressivity near the boundary, which is irrelevant for an eigenfunction that vanishes there.

Appendix A.2. Interval construction

Algorithm 1 Verified or refused certificate for $(\psi, \mathrm{grid}, k^*)$.

1. $\eta \leftarrow \eta(\psi)$, $r \leftarrow H_h\psi - \eta\psi$, $\rho \leftarrow \|r\|_h$. If $\rho = 0$ return the exact certificate.
2. $k \leftarrow N(\eta)$ (verified count). Obtain an approximate spectrum from ARPACK for states $k-1, k, k+1$ only.
3. For `state` in $\{k-1, k\}$: build $a = \hat{\lambda}_{\mathrm{state}-1} + 0.1(\hat{\lambda}_{\mathrm{state}} - \hat{\lambda}_{\mathrm{state}-1})$ and $b = \hat{\lambda}_{\mathrm{state}+1} - 0.1(\hat{\lambda}_{\mathrm{state}+1} - \hat{\lambda}_{\mathrm{state}})$; require $a < \eta < b$ and $N(b) - N(a) = 1$; require $\min(\eta - a, b - \eta) > 10^{-9}s$; keep the candidate with the larger gap.
4. If no candidate verifies, run the count-only bisection fallback and verify its interval in the same way; if that fails, return `valid=False` with the reason.
5. Compute $\gamma = \min(\eta - a, b - \eta)$. If $\rho^2 < (b - \eta)(\eta - a)$ return the Temple enclosure (9); otherwise return the Weinstein interval and mark the Temple bound unavailable. In both cases return $\arcsin(\min(1, \rho/\gamma))$.

Appendix A.3. Reproducibility details

All runs were performed on a laptop CPU (Windows 11, Python 3.11, NumPy 2.4 [23], SciPy 1.17 [24], PyTorch 2.12, the thread count fixed per run in the sweep manifest—four for a one-dimensional run, two for a two-dimensional one). The JSON manifests record the wall-clock time of each job. The reference spectra used for validation are computed by eigenvalue solvers independent of the certificate, applied to the same assembled operator: in one dimension by LAPACK's symmetric tridiagonal solver [24], with the eigenvalue taken as the Rayleigh quotient of the returned vector so that the absolute accuracy is not limited by $\varepsilon\|H_h\|$ (7×10^{-12} to 2×10^{-9} on the three one-dimensional grids); in two dimensions by a dense LAPACK decomposition of the full 6241×6241 matrix (`scripts/dense_reference_2d.py`), which agrees with the ARPACK values to all ten printed digits. ARPACK is used only to propose interval endpoints, never to accept them.

The test suite contains 143 tests and runs with `pytest` or with an included minimal runner when `pytest` is unavailable. The tests that matter for the correctness of the certificate are the ones that compare the counts with dense diagonalization (both dimensions), the ones that assert containment for exact eigenvectors, the ones that assert *refusal* of a degenerate state by the single-state certificate, and (`test_certify_subspace.py`) the ones that certify the same degenerate pair as a two-dimensional subspace, check that the certified angle bound dominates the true principal angle, and check that the certificate is invariant under a change of basis of the trial subspace.

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